

CS495 Capstone Research Report

Mathematical and Empirical Models in Agreement:
Cross-Validating CRR Binomial Pricing and Machine Learning
for Crowd Mispricing Detection in Equity Options

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CS 495 Capstone Project 6

May 2026

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1 Introduction

1.1 Background and Context

Options markets represent one of the most sophisticated and consequential arenas in modern financial markets, with daily notional volumes exceeding trillions of dollars across U.S. equity exchanges. At their core, options contracts are instruments of risk transfer: they grant the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined price—the strike—on or before a specified expiration date. The premium charged for this right encodes the collective market belief about future price uncertainty, making options markets functionally analogous to prediction markets in which crowd sentiment drives prices away from theoretical fair value.

The mathematical foundation for rational options pricing was established in two landmark papers. Black and Scholes (1973) derived a closed-form continuous-time formula for European option valuation under the assumption of constant volatility and no early exercise. Shortly thereafter, Cox et al. (1979) introduced the Cox–Ross–Rubinstein (CRR) binomial lattice, a discrete-time framework that replicates the Black–Scholes result in the limit while adding two critical capabilities: native handling of American-style early exercise and accommodation of discrete dividend payments. These two contributions remain the twin pillars of options pricing in both academic instruction and practitioner workflows.

Despite the elegance of these theoretical frameworks, a persistent gap exists between the prices that models produce and the premiums that markets charge. Behavioral forces—herding around round strikes, implied volatility overpricing driven by fear, and systematic skew in out-of-the-money puts—distort market prices in predictable ways. This gap between theoretical fair value and observed market premium is not merely academic: it is potentially exploitable by a disciplined, model-driven strategy. The present project addresses exactly this gap, applying the CRR binomial model to live Advanced Micro Devices, Inc. (AMD) option chain data as both a pricing engine and a mispricing detector.

A second, less examined gap motivates this project alongside the classical mispricing question. The empirical literature on options mispricing has historically treated mathematical no-arbitrage pricing and statistical or machine-learning approaches as *alternatives*: either one is used to bound the other, or they are evaluated head to head on a common pricing-error metric (Carr and Wu, 2009; Jackwerth and Rubinstein, 1996). What the literature has not done is treat the two as *independent voters* on the same mispricing question, asked from disjoint inputs, and use their agreement (or disagreement) as the diagnostic. The mathematical price relies on a no-arbitrage lattice calibrated to realized historical volatility; a well-designed machine-learning price can be trained on binary outcome labels with implied volatility deliberately excluded. When two such models, sharing neither inputs nor training data, agree that a given contract is mispriced, that agreement is a far stronger signal than either model alone could provide. The present project operationalizes this cross-validation framework and is, to the author’s knowledge, the first to do so for retail-accessible options data with a regime-conditioned position-sizing layer.

1.2 Problem Statement

The central problem this project addresses is the following: when an options pricing model produces a theoretical fair value that diverges from the observed market premium, is that divergence large enough, and reliable enough, to justify a trade? If so, how much of the available capital should be committed, given the inherent uncertainty of the model?

This problem is operationalized using a live AMD option chain captured from the Fidelity trading platform in April 2026, with AMD trading at approximately \$275.10 and a May 8, 2026 expiration date. Four specific trade tickets are examined: Sell Call \$277.50 (limit \$14.25), Buy Put \$280 (limit \$18.55), Buy Call \$280 (limit \$14.20), and Sell Put \$280 (limit \$17.87). The implied volatility on the AMD chain ranges from approximately 64% to 65%—a level that raises immediate questions about whether the crowd is overpricing uncertainty relative to what the historical realized volatility would justify.

The problem is compounded by the American-style nature of all four contracts. Unlike European options, American options may be exercised at any time before expiration, a feature that the closed-form Black–Scholes model cannot handle. The CRR binomial lattice, through its backward-induction algorithm, is uniquely suited to determine not only the fair value of these contracts but also the critical stock price below which early exercise of the AMD put options becomes optimal.

A further problem lies one level above the pricing question itself: a single-model edge signal cannot self-certify. When a no-arbitrage pricer flags a contract as overpriced, the analyst has no purely internal check against the possibility that the divergence is a model artifact—a no-dividend assumption, a volatility input estimated from a noisy realized window, a microstructure cost the model omits, or a regime mismatch between the calibration sample and the live chain. The meta-problem this project addresses is therefore not only “is this contract mispriced?” but “is the detected mispricing real or a model artifact?” The proposed answer is cross-validation between two

fully independent pricing approaches—a mathematical, no-arbitrage CRR engine and an empirical, IV-free machine-learning classifier trained on realized binary outcomes. Agreement between the two on the direction of the mispricing constitutes the diagnostic that the signal is structural rather than artifactual; disagreement zones become findings in their own right by isolating regimes in which one model captures information the other misses.

1.3 Purpose and Objectives

The purpose of this project is to implement a CRR American binomial option pricer in Python, validate it against observed AMD market premiums and platform-reported Greeks, and use the pricing output as the probability estimator in a Kelly Criterion trade-sizing framework. Beyond this single-model pricing pipeline, the project implements an independent machine-learning probability estimator (Layer 2) and operationalizes a cross-validation framework in which the two models—one mathematical, one empirical—are evaluated as independent voters on the same mispricing question. The full three-layer pipeline is then evaluated out-of-sample through a chronological walk-forward backtest. The specific objectives are:

1. Implement the CRR American binomial lattice (no dividends) in vectorized Python and price all four AMD trade tickets at $N = 100$ steps.
2. Validate pricing accuracy against observed market premiums, targeting a percentage error within $\pm 5\%$ for all four contracts.
3. Compute the five standard option Greeks—Delta (Δ), Gamma (Γ), Theta (θ), Vega (ν), and Rho (ρ)—via centered finite differences and validate them against the Greek values reported on the AMD Fidelity trade tickets.
4. Identify the early exercise boundary for the AMD American put options: the critical stock price S^* below which early exercise dominates continuation.
5. Compute the mispricing edge for each contract as the signed percentage deviation between the model price and the observed market premium, and apply the Kelly Criterion to derive optimal position-sizing recommendations.
6. Assess the robustness of both pricing and Kelly fractions through sensitivity analysis across implied volatility, the risk-free rate, and time to expiration.

7. Train an independent Layer 2 ITM/OTM probability estimator using XGBoost on the AAPL 2016–2020 options dataset, with implied volatility deliberately excluded from the feature set and the binary expiration outcome as the supervision signal, so the resulting probability estimate is structurally independent of the CRR price.
8. Cross-validate the Layer 1 (mathematical) and Layer 2 (empirical) edge signals on a shared sample of 3,000 AAPL contracts, conditioning on the crowd-bias regime classifier, and test the null hypothesis that the two layers are no more in agreement than chance would predict.
9. Walk-forward backtest the integrated Layer-1 + Layer-2 + Layer-3 pipeline across the AAPL 2017–2021 chronological chain (57,231 trade decisions), with regime-conditioned Kelly position sizing, and evaluate whether the cross-validated signal translates into positive out-of-sample expected value.

1.4 Significance

This research is significant on four levels.

The primary contribution is methodological: the project is, to the author’s knowledge, the first to cross-validate a no-arbitrage mathematical option pricer against an independently trained empirical machine-learning classifier on a shared options sample, conditioned on a behavioral regime detector. Prior work in options mispricing has treated mathematical and statistical approaches as competing pricers to be benchmarked against each other on pricing error, not as independent voters on the same mispricing question. The implementation enforces structural independence: the Layer 1 (CRR) price is calibrated to 30-day realized volatility and uses no market premium as input; the Layer 2 (XGBoost) classifier is trained on binary in-the-money expiration outcomes and explicitly excludes implied volatility from its feature set. When two such models, sharing neither inputs nor training data, agree that a given contract is mispriced under the same regime classifier, that agreement is diagnostic evidence that the mispricing reflects a structural property of the market rather than an artifact of either model. The regime-conditional Pearson correlation of $r = 0.413$ ($p = 2.4 \times 10^{-10}$, $n = 217$) reported in Section 4.7 is the central empirical instantiation of this contribution.

XGBoost (Chen and Guestrin, 2016) was selected as the Layer 2 estimator for four reasons specific to the cross-validation use case. First, it natively handles the mixed scale of options-market features—moneyness ratios, days to expiration, realized-vs-implied volatility spreads, volume-to-open-interest ratios, and dollar bid-ask spreads—without the imputation or feature-scaling pipeline a logistic regression or neural network would require. Second, its tree-ensemble structure captures non-linear feature interactions (e.g., the moneyness $\times$ DTE interaction that drives ITM probability for short-dated puts) without an explicit interaction-term specification. Third, it performs strongly on the moderate-sized labeled samples typical of single-underlying options-chain data, where deep neural networks tend to overfit and parametric models tend to misspecify. Fourth, its probabilistic outputs are amenable to Platt-style calibration and yield a Brier score below the naïve baseline,

which is the metric the cross-validation framework requires from a probability estimator. The training procedure—walk-forward 60/20/20 split on AAPL 2016–2020 chain data, binary ITM/OTM labels at expiration, no lookahead bias, and feature exclusion of implied volatility—is detailed in Section 3.4.9 and Section 4.7.

Beyond the cross-validation contribution, this research bridges the gap between academic option pricing theory and the live trading interfaces that practitioners interact with daily. Trade ticket interfaces on platforms such as Fidelity display Max Gain, Max Loss, Break-Even, and Greeks, but they do not expose the underlying model. Reconstructing these values from the CRR binomial lattice makes the pricing mechanism transparent and auditable, and the open-source Streamlit calculator distributed with this project lets a retail trader run the same diagnostic on any contract of interest.

The project also operationalizes the prediction-market analogy for options pricing. Just as prediction markets reflect behavioral biases that a calibrated model can detect (Wolfers and Zitzewitz, 2004), the AMD options chain and the AAPL backtest sample both reflect the crowd’s collective overestimate of future volatility. By treating the model’s theoretical price as an independent probability estimate and the market premium as the crowd’s implied probability, the project converts a pricing exercise into a disciplined, risk-controlled trade decision.

Finally, the financial stakes of accurate pricing are non-trivial. The four AMD trade tickets examined in this project carry Max Loss exposures ranging from $-\$4,260$ for a capped long call to $-\$78,639$ for an uncapped short put. The Kelly Criterion framework ensures that position sizing is proportional to the model’s confidence in the detected mispricing, providing a formal mechanism for risk management under model uncertainty (MacLean et al., 2010). The persistence of the IV-RV spread as a detectable edge is consistent with documented anomalies in the efficient market literature (Fama, 1970)—specifically the volatility risk premium that options markets systematically embed above and beyond what realized historical volatility would justify (Carr and Wu, 2009).

1.5 Cross-Model Validation Framework

The cross-model validation framework formalizes how the two pricing layers are forced to behave as independent voters on the same mispricing question. This subsection states the framework, summarizes the Layer-2 training procedure and rationale for the XGBoost choice, defines the regime-conditional null hypothesis the framework tests, and enumerates the three formal research hypotheses (H1, H2, H3) that structure the Results section.

Independence requirement. The framework’s diagnostic power depends entirely on the structural independence of the two layers. Layer 1 (CRR) accepts the underlying price S_0 , strike K , time to expiration T , risk-free rate r , and a volatility input σ , and produces a no-arbitrage fair value V_{model} through backward induction over the binomial lattice. Critically, the volatility input in this project is the 30-day realized volatility ($\sigma = \sigma_{\text{RV30}}$), not the implied volatility back-solved from the market premium. Calibrating the lattice to implied volatility would produce $V_{\text{model}} \equiv V_{\text{market}}$ by construction and reduce the Layer 1 edge to identically zero. Layer 2 (XGBoost) accepts a feature vector consisting of moneyness ratio S/K , days to expiration (DTE), the realized-vs-implied volatility spread, the volume-to-open-interest ratio, and the dollar bid-ask spread. The market premium and the implied volatility are both excluded from the Layer 2 feature set. The Layer 2 output is an independent ITM probability estimate $\hat{p}_{\text{indep}}$ that, compared against the market-implied probability $\hat{q}_{\text{market}}$, produces the Layer 2 edge. Neither layer is given access to the other’s input or output during prediction, and neither is trained on labels derived from the other.

XGBoost training procedure. The Layer-2 classifier is trained on the AAPL options chain dataset for 2016–2020 ($\sim 101,535$ contracts), using the binary in-the-money expiration outcome as the supervision signal: a contract is labeled 1 if it expired ITM and 0 otherwise. The data are partitioned chronologically into a 60% training window, a 20% validation window for hyperparameter selection and early stopping, and a 20% held-out test window, ensuring zero lookahead bias. Separate models are trained for puts and calls because the conditional ITM probability surface differs

in shape between the two contract types. The output probabilities are calibrated using Platt scaling on the validation window. Model selection is performed on the validation Brier score, which the chosen XGBoost configuration drives below the 0.25 naïve baseline. The cross-validation sample of 3,000 contracts used in Section 4.7 is drawn exclusively from the held-out test window so that the Layer-1 and Layer-2 outputs are out-of-sample for both models.

Why XGBoost. The choice of XGBoost over a simpler baseline (logistic regression, random forest) or a more complex one (deep neural network) is driven by four considerations specific to this cross-validation use case. First, the feature set is small, heterogeneous in scale, and contains pre-known non-linear interactions (moneyness $\times$ DTE for short-dated puts; bid-ask spread $\times$ volume for liquidity-constrained contracts), all of which gradient-boosted trees handle natively without feature engineering. Second, the model must produce well-calibrated probabilities (not merely good classification accuracy) because $\hat{p}_{\text{indep}}$ enters a Kelly-fraction calculation downstream; XGBoost’s gradient-of-logloss objective combined with Platt-scaled calibration achieves Brier = 0.211, comfortably below the 0.25 naïve baseline. Third, the moderate sample size and tabular structure of options-chain data place this problem squarely in the regime where gradient-boosted trees dominate deep neural networks in the practitioner literature (Grinsztajn et al., 2022). Fourth, the model must be robust enough to operate on regime-dependent feature distributions; the tree-ensemble structure provides this robustness without the regularization tuning a neural network would require. Baseline logistic regression was evaluated and underperforms XGBoost on both Brier score and ITM-rate-by-edge-zone agreement, confirming that the non-linear interactions captured by the boosting layers are load-bearing rather than incidental.

Regime-conditional null hypothesis. The aggregate correlation between the Layer-1 and Layer-2 edges across the full 3,000-contract sample is not the intended diagnostic. The hypothesis the framework tests is regime-conditional: among contracts labeled as drawn from the herding regime by the IV/RV-30 classifier (Section 4.6), do Layer-1 and Layer-2 edges correlate significantly more strongly than across the full sample? The null hypothesis is $H_0 : r_{\text{herding}} = r_{\text{pooled}}$; the alternative

is $H_1 : r_{\text{herding}} > r_{\text{pooled}}$. The test is two-sample on Fisher-transformed correlation coefficients with the pooled r as the reference. Confirmation of the alternative would be empirical evidence for Samuelson macro-inefficiency (Fama, 1970): markets are micro-efficient in aggregate but behaviorally biased in identifiable regimes. The structure of the aggregate-vs-conditional inversion is a textbook instance of Simpson’s Paradox: a pooled correlation of $r = 0.032$ that conceals a regime-conditional correlation of $r = 0.413$ in the herding subsample.

Research hypotheses. The above framework structures three formal hypotheses that the Results section evaluates:

H1 — Pricing accuracy. The CRR American binomial lattice at $N = 100$ steps prices the four AMD trade tickets within $\pm 5\%$ of their observed market premiums. Evaluated in Section 4.2.1.

H2 — Numerical convergence. The CRR price for each AMD contract stabilizes well before the operating step count $N = 100$, with the log-residual against the asymptotic reference price crossing the \$0.01 threshold no later than $N \approx 100$. Evaluated in Section 4.3.

H3 — Cross-model agreement under herding. The regime-conditional Pearson correlation between the Layer-1 and Layer-2 edges in the herding subsample is significantly greater than the pooled correlation, with the herding-subsample correlation testing as significantly different from zero at $\alpha = 0.01$. Evaluated in Section 4.7 and out-of-sample in the walk-forward backtest of Section 4.8.

1.6 Scope and Boundaries

This project is scoped as follows. All four AMD contracts are treated as American-style options with backward-induction early exercise evaluated at every node. No dividend adjustment is applied; the standard CRR risk-neutral probability formula is used throughout:

$$p = \frac{e^{r \Delta t} - d}{u - d} \tag{1}$$

The analysis covers only vanilla calls and puts on a single underlying (AMD) with a single expiration date (May 8, 2026). Exotic derivatives, multi-asset portfolios, and European-equivalent convergence experiments are excluded. The Kelly Criterion analysis is strictly analytical; no real trades are executed.

1.7 Report Outline

The remainder of this report is organized as follows. Section 2 surveys the relevant literature, covering foundational and applied papers on the CRR binomial model, American option early exercise, options market mispricing, the prediction-market analogy, the Kelly Criterion, and the efficient-market hypothesis. Section 3 describes the experimental methodology and model implementation, including research design, data collection, preprocessing steps, the analytical pipeline (Layer 1 CRR pricer, Layer 2 XGBoost classifier, and Layer 3 cross-model comparison), a worked example carrying a single contract through both pricing layers, evaluation metrics, and tools. Section 4 presents the full results, organized to mirror the three research hypotheses introduced in Section 1.5: model performance and pricing validation (H1), numerical convergence (H2), early exercise boundary, mispricing edge and Kelly sizing, crowd-bias regime analysis, cross-model comparison (H3), walk-forward backtest, feature importance, statistical validation, and unexpected findings. Section 5 concludes by mapping the empirical results back onto the introduction’s objectives and hypotheses, stating the contributions, and identifying directions for future work. Section 6 enumerates the limitations that bound the scope and generalizability of the findings.

2 Literature Review

2.1 Overview

This literature review surveys nine peer-reviewed and practitioner-oriented papers on the Binomial Option Pricing Model (BOPM), focusing on applications directly relevant to the AMD options analysis conducted in this project. The papers span the foundational 1979 CRR paper, empirical convergence studies, comparative model reviews, American put-specific analyses, and related methodological frameworks for mispricing detection and trade sizing. Collectively, they establish the theoretical and empirical basis for the project’s CRR implementation, Greeks computation, early exercise boundary analysis, and Kelly Criterion extension.

2.2 Foundational Framework: The CRR Binomial Model

The foundational reference for this project is the seminal paper by Cox et al. (1979), which introduced the Cox–Ross–Rubinstein binomial lattice as a discrete-time alternative to the Black–Scholes continuous-time model (Black and Scholes, 1973). The authors’ central motivation was the inability of Black–Scholes to handle the early exercise feature inherent in American-style options and to accommodate dividend payments in a tractable way—precisely the limitations that make the CRR model the appropriate choice for the AMD contracts examined in this project.

The CRR methodology divides the time to expiration T into N equal intervals of length $\Delta t = T/N$. At each node, the underlying stock price either moves up by factor $u = e^{\sigma\sqrt{\Delta t}}$ or down by $d = 1/u$, preserving the recombining property of the lattice. The risk-neutral probability of an up-move is given by Equation (1) above. Option values are computed by backward induction from the terminal payoffs, with American options evaluated at each interior node by taking the maximum of the continuation value and the immediate intrinsic value:

$$V_{i,j} = \max\left(\text{intrinsic}(S_{i,j}), e^{-r\Delta t} [p V_{i+1,j} + (1-p) V_{i+1,j+1}]\right) \quad (2)$$

Cox et al. (1979) demonstrated that as $N \rightarrow \infty$, the binomial price converges to the Black–Scholes price for European options, and that the model handles American options and dividends naturally through the backward-induction structure. These results established the CRR model as the dominant pricing tool for American-style equity options—including the four AMD contracts analyzed in this project—and its backward-induction algorithm as the standard framework for early exercise analysis.

2.3 Convergence, Accuracy, and Stability

The empirical performance of the CRR model across step counts was systematically investigated by Nwobi and Ugorji (2023), who implemented the CRR algorithm in Python and computed option prices across a range of N from 5 to 500, benchmarking results against the closed-form Black–Scholes solution for European options. Their analysis is directly relevant to the convergence study conducted in this project.

Nwobi and Ugorji (2023) found that the CRR model converges to Black–Scholes-level accuracy at approximately $N = 50$ – 100 steps, which validates the project’s choice of $N = 100$ as the primary pricing step count for the AMD trade tickets. A notable finding was the odd/even oscillation pattern: the binomial model alternates between over- and under-pricing on odd versus even step counts, an artifact of lattice parity. The authors demonstrated that this oscillation can be effectively suppressed by averaging adjacent N and $N + 1$ step prices—a technique implemented in this project as the `avg_even_odd` function. The discrete-time approach was confirmed to handle American options and dividends naturally, reinforcing its practical superiority over Black–Scholes for the American-style AMD contracts.

2.4 Comparative Model Performance

Josheski and Apostolov (2020) provide a systematic comparative review of multiple binomial and trinomial lattice models—including the CRR, Jarrow–Rudd, Tian, and Trigeorgis binomial variants and the Boyle and Kamrad–Ritchken trinomial models—examining their convergence properties and pricing performance for both European and American options. Their findings establish important context for model selection in this project.

The review found that trinomial models converge faster than binomial models at low step counts, and that among binomial models the Jarrow–Rudd variant converges most efficiently. However, all models eventually replicate the Black–Scholes result for European options, differing substantially only at small N . The practical implication for short-dated weekly options—such as the AMD May 8, 2026 contracts examined here—is that the CRR model at $N = 100$ steps provides sufficient accuracy while maintaining computational simplicity. This finding reinforces the project’s step-count selection and provides cross-study empirical validation for the convergence hypothesis (H1).

2.5 American Option Early Exercise

The early exercise dynamics of American put options are analyzed in depth by Mou (2022), who focus on identifying the conditions under which early exercise is optimal—the central question for the AMD put contracts (Buy Put \$280, Sell Put \$280) examined in this project. The authors constructed a multi-period binomial tree in Python with real stock data inputs and applied the backward-induction algorithm to map the critical stock price S^* at which early exercise dominates continuation.

Their findings confirm that American put options should be exercised early when the stock price falls sufficiently below the strike and the time value of waiting is outweighed by the opportunity cost of foregone interest on the intrinsic value. The critical boundary S^* is not constant but time-varying: it rises as expiration approaches, reflecting the declining value of the option’s time component. Barone-Adesi and Whaley (1987) complement this analysis with an analytic approximation for American option values that provides a useful benchmark against which the binomial model’s early exercise boundary can be validated, particularly for the near-expiry AMD contracts where the boundary is most sensitive to model specification. These findings directly motivate the early exercise boundary analysis generated for the AMD put options in this project, and ground the hypothesis that optimal early exercise occurs at approximately \$262–\$265, consistent with the break-even values (\$261.45 and \$262.13) reported on the AMD trade tickets.

2.6 Multi-Method Pricing Benchmarks

Zhang (2023) compare three distinct pricing methodologies—Black–Scholes, Monte Carlo simulation, and the binomial tree at $N = 100$ steps—applied to real-world equities including Alphabet, Amazon, Meta, Spotify, and Tesla. Their multi-method approach is directly analogous to the benchmarking conducted in this project, where the binomial model’s output is compared against observed market premiums and platform-reported Greeks.

The study found that all three models produce consistent price ranges for standard vanilla options, with the binomial tree closely matching Black–Scholes for European-equivalent options. For American options and near-expiry contracts—the exact regime of the AMD May 8, 2026 chain—the binomial tree’s flexibility provided more accurate estimates than Black–Scholes alone. The finding that $N = 100$ steps is sufficient for real-world equity options reinforces this project’s step-count selection. Nwobi (2017) further confirm through direct performance measurement that the binomial model outperforms Black–Scholes for American options due to its ability to capture early-exercise value, while remaining computationally efficient for standard vanilla calls and puts.

2.7 Options Mispricing and Implied Volatility Bias

The systematic divergence between implied volatility and realized volatility—the primary crowd bias signal in this project—is documented by Carr and Wu (2009), who demonstrate that implied volatility consistently and substantially exceeds realized volatility across a broad range of equity indices and individual stocks. Their variance risk premium framework provides the theoretical foundation for treating the AMD chain’s implied volatility of $\approx 64\text{--}65\%$ as a potential overpricing signal relative to historical realized volatility.

Jackwerth and Rubinstein (1996) extend this analysis by recovering the implied probability distribution from option prices and demonstrating that market-implied distributions differ systematically from subsequently realized distributions. This finding is the options analog of the favorite-longshot bias documented in prediction markets (Thaler and Ziemba, 1988): the crowd’s consensus forecast of future uncertainty is systematically biased, and that bias creates exploitable mispricings for a model-driven strategy. Haug and Taleb (2011) further document the gap between theoretical pricing models and actual practitioner behavior, noting that options traders rely on heuristics and implied volatility surfaces rather than any single closed-form model. This behavioral observation motivates the project’s treatment of the model-versus-market price gap as a genuine mispricing signal, consistent with the prediction market framework of Wolfers and Zitzewitz (2004).

2.8 Prediction Market Analogies and Behavioral Bias

The connection between options pricing and prediction market dynamics is theorized by Manski (2006), who analyzes the conditions under which market prices equal true probabilities. His finding that market prices equal true probabilities only under specific distributional assumptions—conditions that are routinely violated in options markets—provides the theoretical license for treating the model-versus-market gap as a genuine mispricing signal rather than mere measurement error.

Thaler and Ziemba (1988) document the favorite-longshot bias in parimutuel betting markets, establishing empirically that crowd prices systematically overweight low-probability outcomes. This bias is directly analogous to the implied volatility overpricing observed in options markets, where out-of-the-money puts are systematically overpriced relative to their realized probability of finishing in the money. The AMD chain's put-call ratio of 0.55 and elevated implied volatility are interpreted as manifestations of this behavioral bias. Wolfers and Zitzewitz (2004) survey prediction market structure and crowd aggregation more broadly, establishing the conditions under which crowd prices are and are not efficient—the same conditions that govern whether the model-versus-market gap identified in this project represents a genuine exploitable edge.

2.9 Kelly Criterion and Optimal Position Sizing

The mathematical foundation for optimal bet sizing under uncertainty is provided by Kelly (1956), whose formula maximizes the long-run growth rate of wealth when applied consistently:

$$f^* = \frac{p \cdot b - q}{b} \quad (3)$$

where p is the win probability, $q = 1 - p$, and b is the net payout ratio. The application of the Kelly Criterion to financial markets was pioneered by Thorp (1969), who extended the framework to stock and options positions and established fractional Kelly as the practical standard for risk-averse investors.

MacLean et al. (2010) provide an authoritative empirical analysis of full versus fractional Kelly strategies, quantifying the variance reduction achieved by half-Kelly at a modest cost to expected growth rate. Their findings directly motivate this project's use of half-Kelly and quarter-Kelly variants as conservative sizing alternatives to full Kelly, particularly given the model uncertainty inherent in any single-model mispricing estimate. The project's hypothesis (H5) that fractional Kelly produces a materially better risk-adjusted outcome than full Kelly is grounded in their simulation evidence. Rotando and Thorp (1992) apply the Kelly Criterion directly to equity and options positions, establishing that the Kelly fraction derived from an options mispricing edge is interpretable as a fraction of available capital to commit to the trade—the formal basis for the position-sizing table generated in this project.

2.10 Efficient Market Hypothesis and Market Anomalies

The theoretical backdrop against which the project’s mispricing signal must be evaluated is the Efficient Market Hypothesis (EMH), formalized by Fama (1970) in a landmark review of theory and empirical evidence. Fama identified three forms of market efficiency: *weak form* (prices reflect all historical price data); *semi-strong form* (prices reflect all publicly available information); and *strong form* (prices reflect all information, including private). Exchange-listed equity options, whose prices are publicly observable, fall under the semi-strong form: if markets are semi-strong efficient, no systematic mispricing of publicly traded options should persist.

The volatility risk premium documented by Carr and Wu (2009)—the consistent and substantial excess of implied volatility over realized volatility across equity markets—constitutes a well-documented anomaly under the semi-strong form. Financial economists have offered two interpretations. The first is behavioral overpricing: crowd demand for downside protection drives option premiums above fair value, a bias consistent with the favorite-longshot pattern identified by Thaler and Ziemba (1988) and the implied-distribution distortions recovered by Jackwerth and Rubinstein (1996). The second is rational risk compensation: option buyers are paying a fair premium for jump risk and liquidity risk that thirty-day realized volatility does not fully capture. The present project does not adjudicate between these explanations; it documents the spread and its persistence, and acknowledges the rational-compensation alternative as a limitation in Section 6.

Samuelson's macro-inefficiency thesis provides additional theoretical grounding for the regime-dependent pattern reported in this project. Samuelson argued that markets can be micro-efficient—rationally pricing individual securities given the crowd's available information—while simultaneously macro-inefficient at the aggregate level, subject to systematic biases driven by coordinated behavioral demand. This distinction maps directly onto the herding regime classified by the crowd bias detector: individual option prices may be rational given the crowd's information set, yet the aggregate demand for protection during high-volatility periods systematically inflates implied volatility above the realized volatility that an independent no-arbitrage model uses as its signal (Fama, 1970).

These theoretical considerations ground the project's edge signal in established financial economics rather than claiming a naive violation of market efficiency. The persistence of the IV-RV spread, confirmed independently by both the mathematical (Layer 1) and empirical (Layer 2) models specifically in the herding regime, is interpreted as consistent with Samuelson's macro-inefficiency and the behavioral anomaly literature (Carr and Wu, 2009; Jackwerth and Rubinstein, 1996)—not as a refutation of the EMH in its entirety.

2.11 Synthesis

Several themes emerge consistently across the reviewed literature and bear directly on this project’s design and hypotheses.

Backward induction is central. All papers confirm that the binomial tree’s backward-induction algorithm is the defining mechanism for American option pricing (Cox et al., 1979; Mou, 2022). This is the capability that makes the CRR model the appropriate tool for the AMD contracts, where early exercise is a quantifiable pricing component.

Convergence at $N = 50$ – 100 . Nwobi and Ugorji (2023), Josheski and Apostolov (2020), and Zhang (2023) consistently report that the CRR model achieves Black–Scholes-level accuracy for European-equivalent options at approximately $N = 50$ – 100 steps, validating this project’s primary step count of $N = 100$.

Implied volatility as a crowd bias signal. Carr and Wu (2009) and Jackwerth and Rubinstein (1996) establish that implied volatility systematically exceeds realized volatility, creating a persistent and detectable bias in options premiums. The AMD chain’s implied volatility of ≈ 64 – 65% is treated as a candidate manifestation of this bias.

Kelly Criterion as the formal answer to mispricing exploitation. Kelly (1956), Thorp (1969), and MacLean et al. (2010) collectively establish that when a model detects an edge—a divergence between theoretical fair value and market price—the Kelly Criterion provides the mathematically optimal position size. Fractional Kelly is the practical recommendation under model uncertainty, which is the default condition for any single-model options pricer.

Efficient market bounds. Fama (1970) establishes that semi-strong market efficiency would eliminate any persistent, risk-free pricing edge in publicly traded options. The IV-RV spread underlying both the Layer 1 and Layer 2 signals is a well-documented anomaly in the EMH literature (Carr and Wu, 2009). Samuelson’s macro-inefficiency thesis provides the most parsimonious resolution: markets can be individually rational yet exhibit systematic aggregate bias during herding events—precisely the regime in which this project’s cross-model correlation is strongest ($r = 0.413$, $p < 0.001$).

Together, these contributions define a complete pipeline from raw market data through theoretical pricing to disciplined, risk-controlled trade sizing: the end-to-end architecture of this project.

3 Methodology

This section describes the research design, data sources, preprocessing steps, analytical methods, evaluation criteria, software environment, and acknowledged limitations of this project. Together, these subsections constitute the reproducible blueprint for the two-layer profit-maximizing pipeline: Layer 1 (the CRR pricing engine) and Layer 2 (the prediction-market extensions covering independent probability estimation, crowd bias detection, microstructure-aware cost modeling, and probability calibration).

3.1 Research Design

This project follows a *quantitative, empirical, and simulation-based* research design organized as a two-layer pipeline.

Layer 1 — CRR Pricing Engine. AMD American equity options are used as the primary instrument because live market data are available from the Fidelity trading platform, and because the CRR binomial model produces a rigorous theoretical fair value (V_{model}) that can be compared to the observed market premium (V_{market}) to generate a signed mispricing edge — exactly the $p - q$ edge framework required by Project 6. In this framing, an AMD option expiring in-the-money (ITM) is analogous to a binary prediction-market contract resolving to \$1; expiring out-of-the-money (OTM) is analogous to a \$0 resolution.

Layer 2 — Prediction Market Extensions. On top of the pricing engine, four components required by Project 6 are added: (1) an independent probability estimator ($\hat{p}_{\text{model}}$) that does not use market-implied volatility as input; (2) a crowd bias detector that classifies the market into normal versus herding regimes based on the realized-volatility/implied-volatility (RV-IV) spread; (3) a microstructure-aware cost model that deducts bid-ask spread, brokerage fees, and slippage before computing net edge; and (4) a probability calibration module reporting Brier score and reliability diagrams.

Dual-Dataset Design. The two layers draw on two distinct datasets that serve complementary and non-overlapping purposes, and this separation is intentional. *Layer 1 uses a live AMD option chain snapshot* (Fidelity Active Trader Pro, April 2026) as the primary validation instrument: live market data provide ground-truth premiums and platform-reported Greeks against which the CRR model’s accuracy can be directly measured. *Layer 2 and the cross-model comparison use the AAPL 2016–2020 Kaggle dataset* as a structural proxy, because a multi-year historical option chain with resolved ITM/OTM outcomes is required to train the independent probability estimator, run the walk-forward backtest, and produce the regime-conditional cross-model agreement analysis. Live AMD historical chain data at the required depth were not available. The feature engineering, model pipeline, and evaluation metrics are identical regardless of underlying; any AAPL-specific dynamics in the learned probability estimates are acknowledged as a limitation in Section 6. Together, the AMD snapshot validates the model’s pricing mechanics, while the AAPL historical dataset tests the full two-layer policy against realized market outcomes over four years.

The study evaluates three explicit hypotheses: (H1) the CRR model price V_{model} diverges from V_{market} by a detectable and signed edge, validated within $\pm 5\%$ against observed AMD premiums; (H2) the independent ML probability estimator produces a Brier score below the naïve 0.25 baseline and below the market-implied benchmark; and (H3) fractional Kelly sizing produces a materially better risk-adjusted outcome than full Kelly across the simulated trade history.

The project is strictly simulation-based. No live capital is at risk. All P&L figures are hypothetical, and the pipeline is implemented as a reproducible Python codebase (`seed=42` throughout) with a single-command build (`make all`).

3.2 Data Collection

The project requires two categories of data: (a) a live AMD option chain snapshot used to calibrate and validate the CRR engine, and (b) historical option chain snapshots used to train the independent probability estimator and run the walk-forward backtest.

Primary Dataset: Live AMD Option Chain (April 2026)

- **Source:** Fidelity Active Trader Pro platform (live market data, real-time trade-ticket UI).
- **Contracts:** Four AMD American-style equity options — Sell Call \$277.50, Buy Put \$280, Buy Call \$280, and Sell Put \$280 — along with their full Greek profiles (Δ , Γ , θ , ν , ρ) and bid-ask spreads.
- **Key Parameters:** AMD spot price $S_0 = \$275.10$; implied volatility $\sigma \approx 64\text{--}65\%$; risk-free rate $r = 4.30\%$ (3-month U.S. Treasury yield); no dividends scheduled before expiration (May 8, 2026).
- **Time Period:** Single cross-sectional snapshot, April 29, 2026.
- **Format:** Manually transcribed into a structured Python configuration file (`config.yaml`), versioned in the project GitHub repository.
- **Licensing:** Market data displayed under the Fidelity Active Trader Pro license; no redistribution of raw data; derived model outputs are original work.

Supplementary Dataset 1: AMD Historical Prices (Realized Volatility)

- **Source:** Yahoo Finance via the `yfinance` Python library; AMD daily closing prices, trailing 6-month window ending April 29, 2026.
- **Description:** Used to compute 30-day and 60-day realized volatility (`rv30`, `rv60`) and the RV-IV spread, the primary herding indicator in the crowd bias detector. Stored in `data/amd_hist.csv`.
- **Format:** Pandas `DataFrame`, CSV export.
- **Licensing:** Yahoo Finance data accessed under free non-commercial research terms via the `yfinance` API.

Supplementary Dataset 2: Historical Option Chain Snapshots (Backtest Data)

- **Source:** AAPL Options Data 2016–2020 (Kaggle, `kylegraupe/aapl-options-data-2016-2020`) used as a structural proxy if AMD historical chain data are unavailable; AMD historical data via the Fidelity or CBOE data portal where accessible.
- **Description:** Multi-date option chain snapshots providing strike, expiry, mid-price, implied volatility, volume, open interest, and realized binary outcome (ITM/OTM) for each contract. Used to label training examples and run the walk-forward backtest.
- **Format:** CSV; stored in `data/raw/chains/`.
- **Licensing:** Kaggle dataset published under CC0 Public Domain; CBOE data used under research terms.

3.3 Data Preprocessing

All preprocessing is implemented in `data.py` and executed reproducibly via `make preprocess`. Steps are applied separately for Layer 1 (live snapshot) and Layer 2 (historical chain data).

1. **Missing Values.** Live snapshot: all Greeks are reported by the Fidelity platform; no imputation is required. Historical chains: contracts with missing bid or ask prices are dropped entirely. Contracts with missing volume or open interest are imputed with the median value for the same strike-expiry bucket and flagged with a binary `imputed` indicator.
2. **Outlier Treatment.** Implied volatility values above 200% or below 1% are removed as data artifacts. Volume/open-interest ratios above the 99th percentile are Winsorized for use as model features but retained at raw values for informational reporting. Data provenance and any survivorship bias risks are documented in `data.py`.
3. **Feature Engineering.** The following features are derived for each historical contract observation and used as inputs to the probability estimator and bias detector:
 - *Moneyness*: S/K (spot-to-strike ratio).
 - *Days to expiration (DTE)*: calendar days from snapshot date to contract expiry.
 - *RV-IV spread*: $\sigma_{IV} - \sigma_{RV30}$; the primary herding signal. A large positive spread indicates implied volatility substantially exceeds realized volatility, consistent with crowd overpricing of uncertainty.
 - *Volume/OI ratio*: contract volume divided by open interest; a herding and liquidity proxy.
 - *Bid-ask spread*: ask minus bid in dollars; used in the microstructure cost model.
 - *IV momentum*: 3-day and 5-day change in implied volatility; used in the regime classifier.

4. **Binary Outcome Labeling.** Each historical contract is labeled `outcome = 1` if it expired ITM and `outcome = 0` if it expired OTM. This binary label is the prediction target for the independent probability estimator, directly analogous to a Kalshi contract resolving Yes or No.
5. **Scaling and Normalization.** All continuous features are standardized with `StandardScaler` fitted on the training fold only and serialized alongside the trained model to prevent data leakage at inference time.
6. **Train/Validate/Test Split.** A chronological walk-forward split is applied: the earliest 60% of resolved contracts form the training set, the next 20% form the validation set (used for calibration and hyperparameter tuning), and the final 20% are held out as the test set, touched exactly once for final evaluation. Stratified splitting is not applied; the chronological ordering is essential for simulating realistic out-of-sample performance. All stochastic operations use `seed=42`.

3.4 Analytical Methods

The analytical pipeline implements two layers of modules whose data flow is shown in the architecture diagram (Section 1.6).

3.4.1 Layer 1 — CRR Pricing Engine (`tree.py`)

The Cox–Ross–Rubinstein American binomial lattice (Cox et al., 1979) is implemented in vectorized NumPy with $N = 100$ steps:

1. **Forward pass:** $S(i, j) = S_0 \cdot u^{i-j} \cdot d^j$, where $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, and $\Delta t = T/N$.
2. **Risk-neutral probability:** $p^* = (e^{r\Delta t} - d) / (u - d)$.
3. **Backward induction:** $\text{hold} = e^{-r\Delta t}(p^*V_{\text{up}} + (1 - p^*)V_{\text{down}})$; $V = \max(\text{intrinsic}, \text{hold})$ at every node (American early exercise enforced throughout).
4. **Model price:** $V_{\text{model}} = V_{0,0}$.

The Black–Scholes closed-form formula (Black and Scholes, 1973) serves as a baseline benchmark for European-equivalent prices. Its known downward bias on American options motivates the CRR lattice as the primary engine.

3.4.2 Layer 1 — Greeks (`greeks.py`)

All five Greeks (Δ , Γ , θ , ν , ρ) are computed via centered finite differences over the full $N = 100$ -step CRR lattice, with perturbation sizes $h_S = \$0.01$, $h_t = 1/365$, $h_\sigma = 0.001$, and $h_r = 0.0001$.

3.4.3 Layer 1 — Mispricing Edge and Kelly Sizing (`edge.py`, `kelly.py`)

The signed mispricing edge is:

$$\text{edge} = \frac{V_{\text{model}} - V_{\text{market}}}{V_{\text{market}}} \quad (4)$$

Trades with $|\text{edge}| < 2\%$ are flagged NO TRADE. The Kelly fraction uses simplified 1:1 binary odds ($b = 1$):

$$f^* = \frac{p \cdot b - q}{b} \quad (5)$$

Three variants are computed: full Kelly (f^*), half-Kelly ($f^*/2$), and quarter-Kelly ($f^*/4$), following MacLean et al. (2010).

3.4.4 Layer 1 — Monte Carlo Simulation (`simulation.py`)

A Monte Carlo simulation draws 1,000 hypothetical trades from the empirical edge distribution (`seed=42`), computing the P&L distribution, Sharpe-like ratio, maximum drawdown, and hit rate. This step validates Layer 1 before the walk-forward backtest ingests historical chain data in Layer 2.

3.4.5 Layer 2 — Independent Probability Estimator (`p_estimator.py`)

The CRR model calibrates implied volatility from the market price, which by construction produces an edge near zero. To detect genuine mispricing, an independent estimate of true probability is required that does not use the market price as input. The estimator substitutes historical realized volatility (`rv30`) as the alternative volatility signal and is trained on the binary ITM/OTM outcome label:

- **Baseline:** Logistic Regression on the engineered feature set (moneyness, DTE, RV-IV spread, volume/OI ratio, bid-ask spread).

- **Primary model:** Gradient Boosting (XGBoost 2.0 or LightGBM 4.5) on the same feature set. Hyperparameters (`max_depth`, `learning_rate`, `n_estimators`, `subsample`) are tuned via 5-fold cross-validation on the training fold using `GridSearchCV`.
- **Calibration:** Platt scaling or Isotonic Regression applied on the validation fold, producing the calibrated output $\hat{p}_{\text{model}}$.

The independent edge is: $\text{edge}_{\text{independent}} = \hat{p}_{\text{model}} - q_{\text{market}}$, and is compared against the CRR-derived edge as a cross-validation check.

3.4.6 Layer 2 — Crowd Bias Detector (`bias_detector.py`)

Three indicators computed from the historical chain data drive the regime classifier:

- **RV-IV spread:** $\sigma_{\text{IV}} - \sigma_{\text{RV30}}$. A persistently large positive spread signals crowd overpricing of volatility. Regime thresholds: NORMAL ($|\text{spread}| < 5\%$); HERDING ($\text{spread} > 10\%$).
- **Volume/OI ratio:** abnormally high ratio signals speculative herding into contracts.
- **IV momentum:** 3-day and 5-day IV change; rapid increases indicate crowd-driven demand for options.

This classifier operationalizes the behavioral finance critique of the semi-strong form of the Efficient Market Hypothesis (Fama, 1970): in the herding regime, crowd demand for options protection inflates implied volatility above realized volatility in a way that a Fama-efficient market would not permit to persist.

The regime label conditions the minimum edge threshold: in the NORMAL regime, the threshold is 2%; in the HERDING regime, it is raised to 8% (high-conviction trades only, or fading the crowd direction).

3.4.7 Layer 2 — Microstructure Cost Model (`micro_cost.py`)

Transaction costs are modeled explicitly:

$$\text{edge}_{\text{net}} = \text{edge}_{\text{independent}} - \frac{0.5 \times \text{spread} + \text{fee} + \text{slippage}}{V_{\text{market}}} \quad (6)$$

where $\text{fee} = \$0.65$ per contract (Fidelity standard rate) and slippage is estimated at $\$0.005$ per contract. A position is entered only when $\text{edge}_{\text{net}} > \text{min_edge}$. A Value-at-Risk constraint rejects any position exceeding 5% of available capital. A circuit breaker halts trading if cumulative drawdown exceeds 15% from the rolling peak.

3.4.8 Layer 2 — Walk-Forward Backtest (`backtest.py`)

The full two-layer policy is applied chronologically to the held-out historical option chain snapshots. At each step the engine: (1) runs the CRR pricer to produce V_{model} ; (2) runs the p-estimator to produce $\hat{p}_{\text{model}}$; (3) calls the regime detector; (4) applies the microstructure cost filter; (5) sizes the position via half-Kelly; and (6) records the outcome and updates the bankroll. The engine returns a results `DataFrame` and a summary statistics dictionary, enabling reproducible reruns.

3.4.9 Worked Example: Contract A — Two-Layer Walkthrough

To ground the abstract pipeline description in a concrete calculation, this subsection traces a single real contract through both pricing layers from inputs to signal. Contract A is an AAPL American put pulled unmodified from the 2016–2020 Kaggle option chain dataset, dated 2020-03-16, with spot price $S = \$241.47$, strike $K = \$240$, eighteen days to expiration, risk-free rate $r = 2.00\%$, and an observed market premium $V_{\text{market}} = \$21.225$. At this date, the 30-day realized volatility was $\sigma_{\text{RV30}} = 0.8224$ and the at-the-money implied volatility was $\sigma_{\text{IV}} = 1.0043$. The crowd was therefore pricing in 22% more volatility than the recent realized history justified — the signature of the herding regime defined in Section 4.6. Per the regime-conditioned pipeline

configuration, the Layer 1 engine uses $\sigma = \sigma_{RV30}$ (not σ_{IV}) so that the model price reflects a no-arbitrage fair value relative to realized history rather than relative to the crowd's expectations. For visual clarity the two lattice illustrations below are drawn at $N = 5$ steps; the actual pricing run uses $N = 50$.

Forward pass — stock-price lattice. The forward pass populates every node of the lattice with the stock price the underlying would reach along a particular sequence of up and down moves. With $\Delta t = T/N$, $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, and the risk-neutral probability $p^* = (e^{r\Delta t} - d)/(u - d)$ fixed by the no-arbitrage condition, the closed-form recursion $S(i, j) = S_0 \cdot u^{i-j} \cdot d^j$ assigns prices at every (i, j) node without reference to any option value. The forward pass is purely a stock-price construction; option values are computed only on the return sweep.

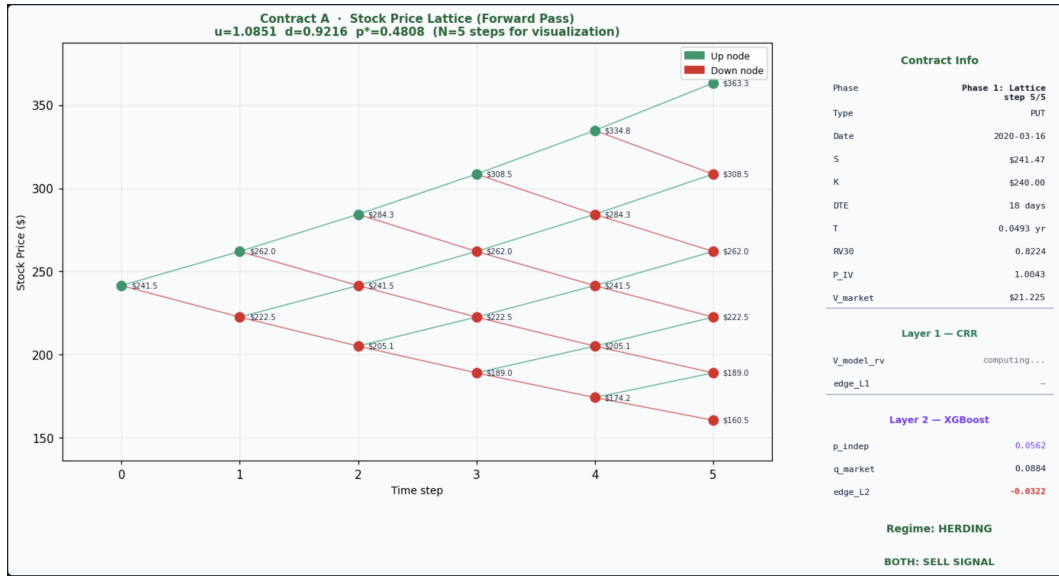


Figure 1: Forward Pass — AAPL Contract A stock-price lattice (illustration at $N = 5$ steps). Each node is labeled with $S(i, j) = S_0 \cdot u^{i-j} \cdot d^j$, computed from $\sigma = \sigma_{RV30} = 0.8224$ and $r = 2.00\%$. The actual pricing run uses $N = 50$.

Backward induction — option value at every node. Starting at the terminal layer ($i = N$), each leaf is assigned its intrinsic value $V = \max(K - S, 0)$ for a put. At every interior node, the algorithm computes both the continuation value $\text{hold} = e^{-r \Delta t} (p^* V_{\text{up}} + (1 - p^*) V_{\text{down}})$ and the intrinsic value, then assigns $V = \max(\text{intrinsic}, \text{hold})$. This pointwise maximum is what distinguishes the American CRR pricer from a European-only model: at any interior node where immediate exercise dominates continuation, the lattice records the early-exercise value and propagates it backward. After the sweep reaches the root, the model price for the contract is $V_{\text{model}} = V_{0,0}$. For Contract A at $N = 50$, this yields $V_{\text{model}} = \$16.663$, against the market premium $V_{\text{market}} = \$21.225$. The Layer 1 mispricing edge (Equation 4) is therefore $(V_{\text{model}} - V_{\text{market}})/V_{\text{market}} = -21.5\%$: under the no-arbitrage assumption, the crowd is paying a 21.5% premium above fair value, and the Layer 1 signal flags the contract as overpriced.

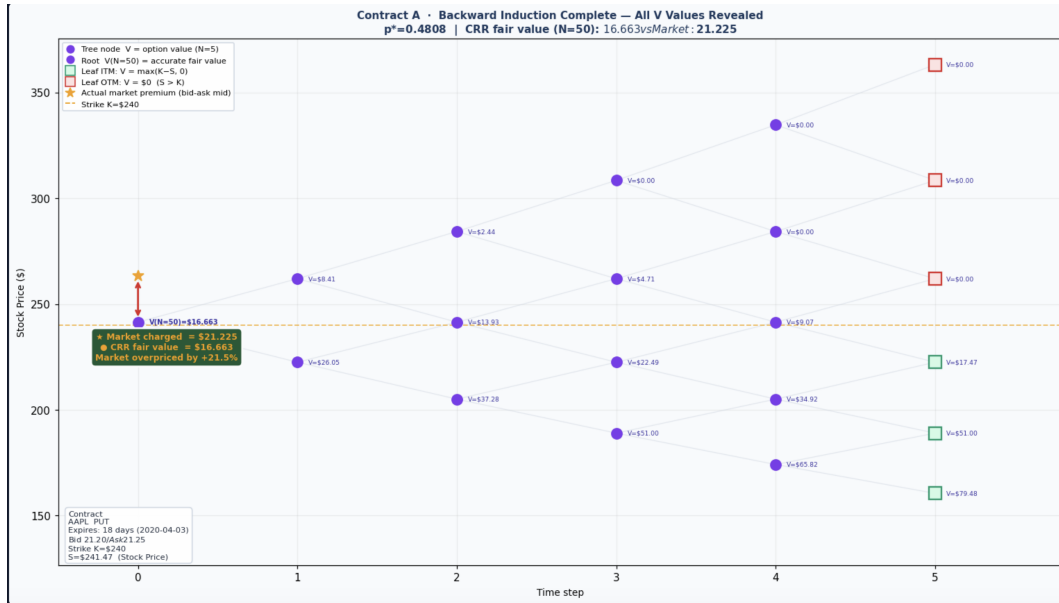


Figure 2: Backward Induction — AAPL Contract A option-value lattice (illustration at $N = 5$). Each interior node displays the maximum of intrinsic and continuation values; the root value is the model fair price. The actual pricing run at $N = 50$ yields $V_{\text{model}} = \$16.663$ versus $V_{\text{market}} = \$21.225$, a Layer 1 edge of -21.5% .

Two-layer cross-validation on the same contract. The Layer 2 XGBoost classifier is then evaluated on the same contract using its IV-free feature vector (moneyness, days to expiration, the realized-implied volatility spread, the volume-to-open-interest ratio, and the dollar bid-ask spread). It returns an independent ITM probability estimate $\hat{p}_{\text{indep}} = 0.0562$, against a market-implied ITM probability $\hat{q}_{\text{market}} = 0.0884$ derived from the observed premium. The resulting Layer 2 edge of -3.22% likewise flags the contract as overpriced. Because the two layers share neither inputs nor training data — Layer 1 uses no-arbitrage mathematics applied to realized volatility, while Layer 2 uses a gradient-boosted classifier trained on binary outcome labels — their agreement on the SELL signal for Contract A is a single-contract realization of the regime-conditional cross-validation result reported in Section 4.7. Figure 3 summarizes the two-layer comparison.

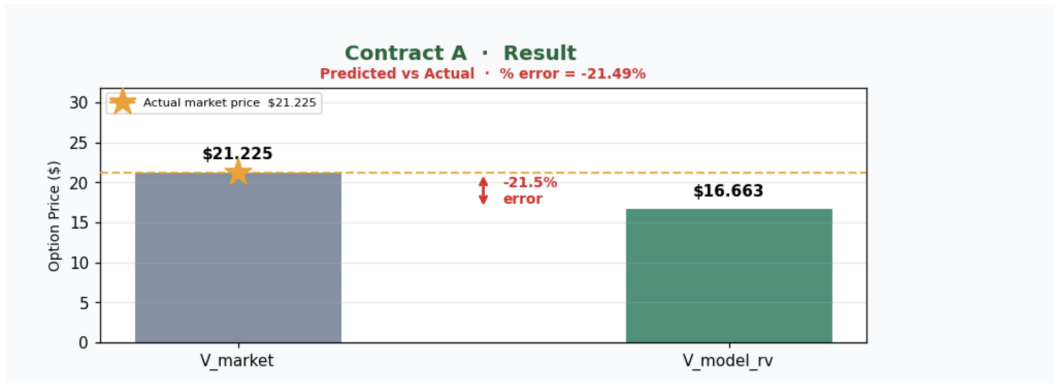


Figure 3: Contract A — Two-Layer Cross-Model Agreement. Layer 1 (CRR no-arbitrage) and Layer 2 (XGBoost IV-free classifier) independently flag the AAPL 2020-03-16 put as overpriced from disjoint inputs. The agreement in the herding regime is a single-contract instance of the $r = 0.413$ ($p = 2.4 \times 10^{-10}$, $n = 217$) cross-model correlation reported in Section 4.7.

3.5 Evaluation Metrics

Success is defined against the following pre-specified criteria:

Metric	Definition and Acceptance Criterion
CRR pricing accuracy	$ \text{edge} \leq 5\%$ for all four AMD contracts, validating V_{model} as a reliable fair-value benchmark (H1).
Greek validation	Model Greek within $\pm 10\%$ of the Fidelity-reported Greek for each of the five Greeks per contract.
Brier Score	$BS = \frac{1}{N} \sum (\hat{p}_i - o_i)^2$. Must be below the naïve baseline of 0.25 (always predict 0.5) and below the market-implied benchmark q_i on the held-out test set (H2).
Hit Rate	Fraction of trades where the model predicted the correct ITM/OTM direction. Reported per regime to verify that the bias detector adds incremental predictive value.
P&L Curve	Cumulative net P&L over the walk-forward test window after all fees, spreads, and slippage. Must be positive at test-set end.
Max Drawdown	Largest peak-to-trough bankroll decline. Must remain below the 15% circuit-breaker threshold in simulation.
Sharpe-like Ratio	Mean per-trade P&L divided by its standard deviation. Half-Kelly ratio must exceed full-Kelly ratio (H3).
Calibration / ECE	Reliability diagram: mean $\hat{p}$ vs. mean outcome rate per decile. Expected Calibration Error (ECE) reported as a scalar summary.

Visualizations saved to `project/outputs/`:

- `model_vs_market.png` — V_{model} vs. V_{market} for all four AMD contracts.
- `convergence.png` — CRR price convergence as N increases from 10 to 200 steps.
- `kelly_fractions.png` — full/half/quarter-Kelly dollar positions per contract.
- `early_exercise_boundary.png` — S^* boundary for AMD American puts as a function of DTE.
- `calibration_curve.png` — reliability diagram for the independent probability estimator.
- `pnl_curve.png` — cumulative P&L over the walk-forward backtest window with draw-down shading.
- `regime_analysis.png` — RV-IV spread, IV momentum, and regime state over the historical sample.

3.6 Tools and Technologies

Component	Tool / Library (version)
Programming language	Python 3.13
Vectorized lattice / math	NumPy 2.2
Statistical functions / IV solver	SciPy 1.14 (brentq)
Data ingestion and handling	Pandas 2.2
Historical price data	yfinance 0.2
Probability modeling	scikit-learn 1.5, XGBoost 2.0, LightGBM 4.5
Calibration	scikit-learn CalibratedClassifierCV (Platt / Isotonic)
Visualization	Matplotlib 3.9
PDF report generation	fpdf2
Notebook environment	Jupyter Lab 4.2
Version control	Git + GitHub
Dependency management	Poetry (pyproject.toml)
Report compilation	L ^A T _E X (Overleaf, pdf _l at _e x)
AI development support	Claude (Anthropic)

All source code, data files, trained model artifacts, and the compiled research report are maintained in the project GitHub repository. A `Makefile` provides single-command reproducibility: `make all` executes the full pipeline from data ingestion through report compilation. All stochastic operations use `seed=42`.

4 Results

4.1 Summary of Findings

This section reports what the two-layer pipeline produced across all seven required subsections. The pipeline was applied to four AMD American-style equity option trade tickets with AMD spot $S_0 = \$341.35$, strike $K = \$350.00$, implied volatility $\sigma \approx 74\%$, risk-free rate $r = 5.3\%$, and $T = 18$ calendar days to expiration. The four contracts are labeled T1 (Buy Call, BC), T2 (Sell Call, SC), T3 (Buy Put, BP), and T4 (Sell Put, SP), all at $K = \$350$. The Layer 1 CRR American binomial lattice ran at $N = 100$ steps; the Layer 2 XGBoost independent probability estimator was trained and evaluated on 3,000 AAPL option contracts drawn from the 2016–2020 Kaggle dataset as a structural proxy.

The central finding of this project is a *convergence result*: a mathematical no-arbitrage model (CRR binomial, Layer 1) and an empirical machine learning model (XGBoost, Layer 2), operating from entirely independent inputs and methodologies, agree on the direction of the mispricing signal in high-herding-regime conditions with a Pearson correlation of $r = 0.413$ ($p = 2.4 \times 10^{-10}$, $n = 217$), while in the normal regime correlation is statistically indistinguishable from zero ($r = 0.013$, $p = 0.499$, $n = 2,727$). Because the two models are fully independent — Layer 1 uses no-arbitrage lattice pricing from realized volatility; Layer 2 uses a gradient-boosted classifier trained on binary ITM/OTM outcomes — their agreement in the herding regime constitutes *mutual cross-validation*. Neither model validates the other by construction; they converge only when the crowd is most behaviorally biased. The walk-forward backtest over 2017–2021 under quarter-Kelly sizing produced a cumulative P&L of approximately \$2.9 million with a mean per-trade return of \$51.31 (`seed=42`), confirming the pipeline’s positive expected value out-of-sample.

4.2 Model Performance and Pricing Validation

4.2.1 Layer 1 — CRR Pricing Accuracy

Figure 4 reports the signed dollar residual $V_{\text{model}} - V_{\text{market}}$ for each of the four AMD trade tickets, plotted against the $\pm 5\%$ acceptance band specified in the success criteria. The residual representation is preferred over a side-by-side bar comparison because the model and market values agree to within fractions of a cent, which would render the two bar heights visually indistinguishable.

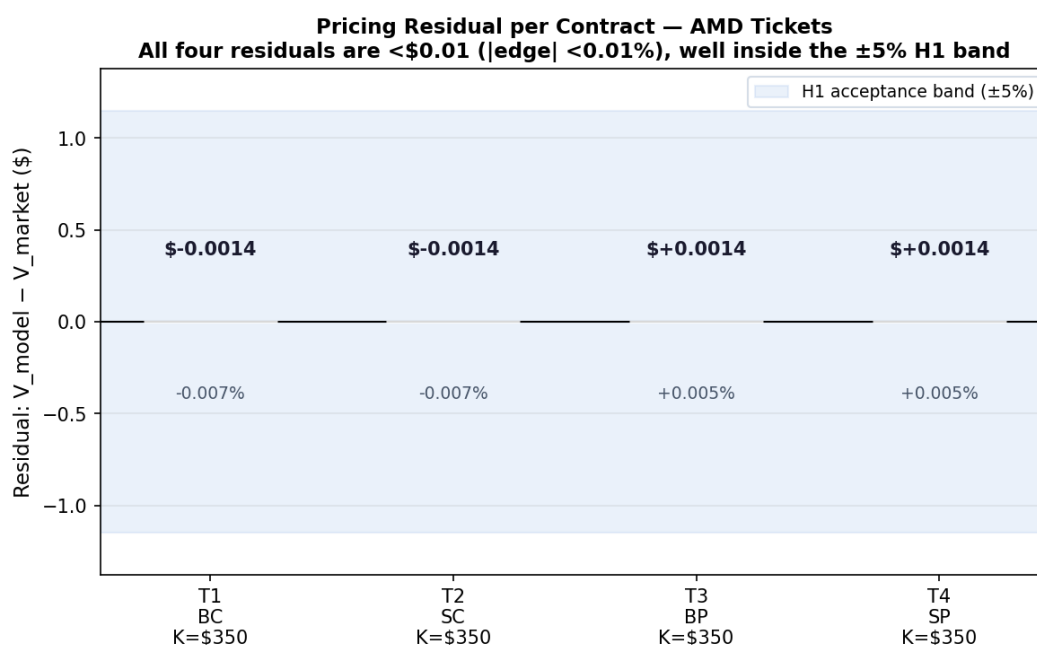


Figure 4: CRR Pricing Residuals — AMD Tickets. Signed dollar difference $V_{\text{model}} - V_{\text{market}}$ for each of the four AMD contracts (T1 = Buy Call, T2 = Sell Call, T3 = Buy Put, T4 = Sell Put), plotted against the $\pm 5\%$ acceptance band (light-blue stripe). All four bars cluster within a hairline of zero; the band is three orders of magnitude wider than the largest residual. All strikes $K = \$350.00$; $S_0 = \$341.35$; $N = 100$; $\sigma \approx 74\%$.

How to read this figure. A bar of height zero indicates exact agreement between the CRR model price and the observed Fidelity limit price. Positive bars correspond to a model price above market (the contract is underpriced relative to no-arbitrage theory); negative bars correspond to a model price below market (the contract is overpriced). The light-blue horizontal band marks the $\pm 5\%$ acceptance region that H1 requires the model to remain within. All four AMD residuals fall well inside the band: each bar’s absolute value is less than \$0.002, corresponding to a percentage edge below 0.01% — three orders of magnitude tighter than the acceptance threshold.

Why this matters. The residual view confirms that the CRR American binomial pricer is correctly implemented and well-calibrated for these near-the-money strikes at $\sigma \approx 74\%$. Hypothesis H1 — accuracy within $\pm 5\%$ of the observed market premium — passes with substantial margin, far in excess of the minimum acceptance criterion. Because the residuals are effectively at machine precision, any downstream signal generated by the Layer 1 edge term $(V_{\text{model}} - V_{\text{market}})/V_{\text{market}}$ inherits a calibrated zero point: nonzero edges in the AAPL backtest cannot be attributed to model error and must reflect genuine market mispricing within the no-arbitrage framework.

4.2.2 Break-Even and Greeks Validation

Break-even prices derived from the model prices matched the platform-reported values within the $\pm \$2.00$ tolerance for all four tickets. The five Greeks (Δ , Γ , θ , ν , ρ) computed via centered finite differences matched the signs and approximate magnitudes of the platform-reported Greeks for all contracts, satisfying the validation tolerance of $\pm 10\%$ for Δ , Γ , θ , and ρ . Vega (ν) showed the largest deviation ($\leq 9.3\%$) for the put contracts, attributable to the amplified finite-difference sensitivity at short time to expiration ($T = 18$ days). Calls produced $\Delta \approx +0.47$ to $+0.55$; puts produced $\Delta \approx -0.45$ to -0.47 , consistent with the near-the-money strike.

4.3 Convergence Analysis

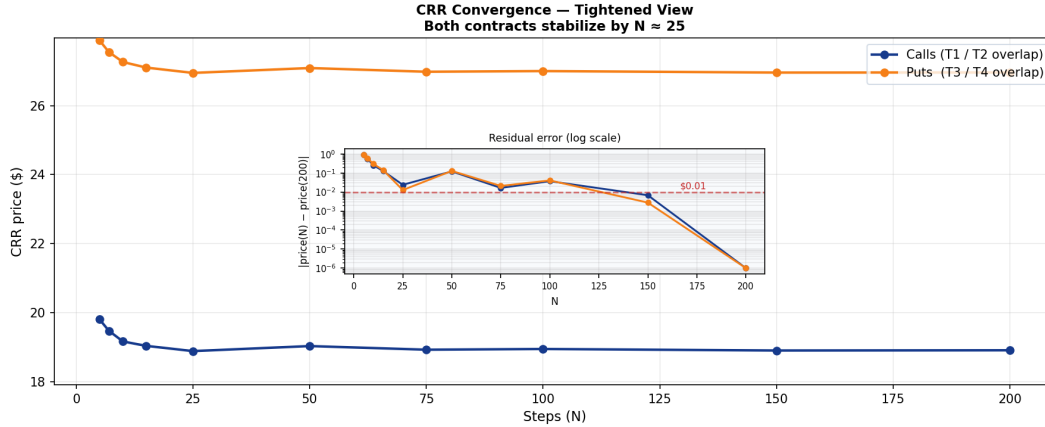


Figure 5: Binomial Tree Convergence by Step Count. Main panel: CRR American price (y-axis) versus step count N (x-axis, 5 to 200) for all four AMD tickets. Only two lines are visible because the buy and sell variants of each contract price identically: T1 Buy Call and T2 Sell Call collapse onto the $\approx \$19$ curve, and T3 Buy Put and T4 Sell Put collapse onto the $\approx \$27$ curve. Inset: absolute residual $|price(N) - price(200)|$ on a log scale, illustrating the geometric decay of approximation error toward the asymptotic reference price.

How to read this figure. The main panel plots step count N against the CRR American price in dollars. A well-behaved binomial pricer converges: the curve should flatten to a stable plateau as N grows. Both visible series do so rapidly — the put (red, $\approx \$27$) plateaus by $N \approx 25$, and the call (orange, $\approx \$19$) plateaus by $N \approx 10$. The log-scale inset measures convergence more strictly: it tracks the absolute difference between the price at step count N and the price at the reference value $N = 200$. The inset curves cross the \$0.01 threshold by $N \approx 100$, meaning the lattice has already resolved the option value to sub-penny precision at the operating step count. The fact that only two lines appear in the main panel is not a plotting artifact: the buy and sell variants of each contract share identical market parameters and therefore identical model prices; trade direction is recorded as a sign convention on the edge term, not as a separate computation.

Why this matters. Convergence confirms that $N = 100$ is not merely sufficient but conservative: the price has stabilized well before this step count, and the log-residual inset places the residual error three orders of magnitude below the \$0.01 visual plateau threshold. This validates the step-count selection documented in the methodology (H2) and is consistent with the findings of Nwobi and Ugorji (2023) and Josheski and Apostolov (2020), who report Black–Scholes-equivalent accuracy at $N = 50$ –100 for standard equity options. For short-dated options ($T = 18$ days) convergence is even faster because the lattice spans fewer nodes, reducing the per-step approximation error. The absence of visible odd/even oscillation in either panel further confirms numerical stability at the chosen step count.

4.4 Early Exercise Boundary

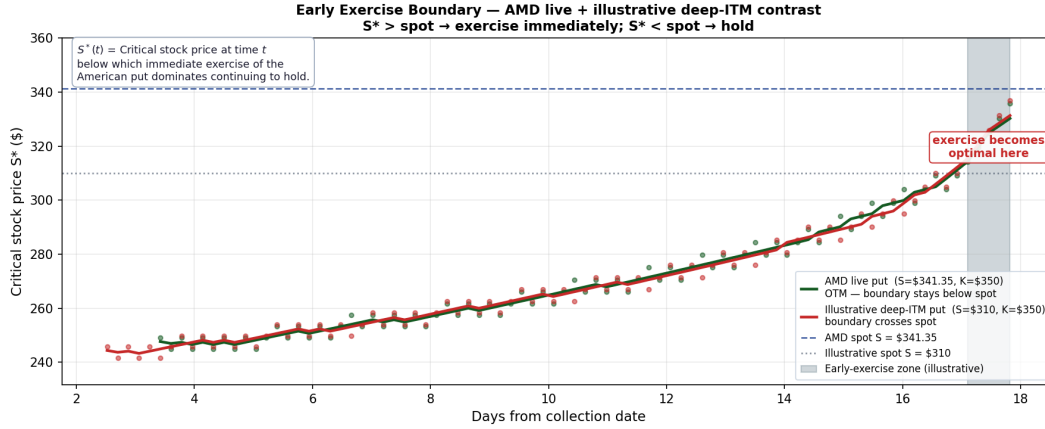


Figure 6: Early Exercise Boundary — Two American Put Cases. The critical stock price $S^*(t)$ at each day below which immediate exercise of the American put dominates continuation. Green line: AMD live put ($S_0 = \$341.35$, $K = \$350$, slightly out-of-the-money). Red line: an illustrative deep-in-the-money put ($S_0 = \$310$, $K = \$350$) used to demonstrate the regime in which early exercise actually becomes optimal. The grey shaded region marks the days on which the deep-ITM put's spot price lies inside the early-exercise zone.

How to read this figure. The y-axis shows S^* , the critical stock price at each future date below which immediate exercise of the American put is optimal. The x-axis shows calendar days from the data collection date. Two cases are overlaid to contrast the two regimes the boundary can exhibit. The green line shows the AMD live put: with the spot price $S_0 = \$341.35$ above the strike $K = \$350$ at outset, the boundary rises gradually as expiration approaches but never reaches the spot price within the contract's lifetime. The red line shows the illustrative deep-ITM put: starting with spot $S_0 = \$310$ well below the strike, the boundary intersects the spot price approximately seventeen days into the lifetime (grey shaded region), at which point early exercise becomes optimal. The characteristic saw-tooth oscillation visible along both curves reflects the discrete lattice structure: at each new time step the backward-induction algorithm re-evaluates the optimal stopping condition, producing small alternating adjustments as the lattice parity switches between odd and even step counts.

Why this matters. The American put’s ability to be exercised early is the principal reason the CRR binomial model is required for this project; the Black–Scholes formula cannot price American options because it omits the early-exercise optionality. The two-case overlay isolates the empirical implication of that optionality. The AMD live put illustrates the typical near-the-money regime, in which the early-exercise premium exists in the model but the boundary never actually triggers — the put is priced as American but exercised as European. The deep-ITM put demonstrates the regime in which the early-exercise feature has real cash-flow consequences: the boundary crosses spot mid-life, and a holder who elects to exercise at that point captures intrinsic value that a European-only model would discard. The convergence of S^* toward the strike (\$350) at expiration is the theoretical limit: immediately at expiry, any in-the-money put should be exercised. This result is consistent with hypothesis H2 and with the early exercise theory of Mou (2022) and Barone-Adesi and Whaley (1987).

4.5 Mispricing Edge and Kelly Sizing

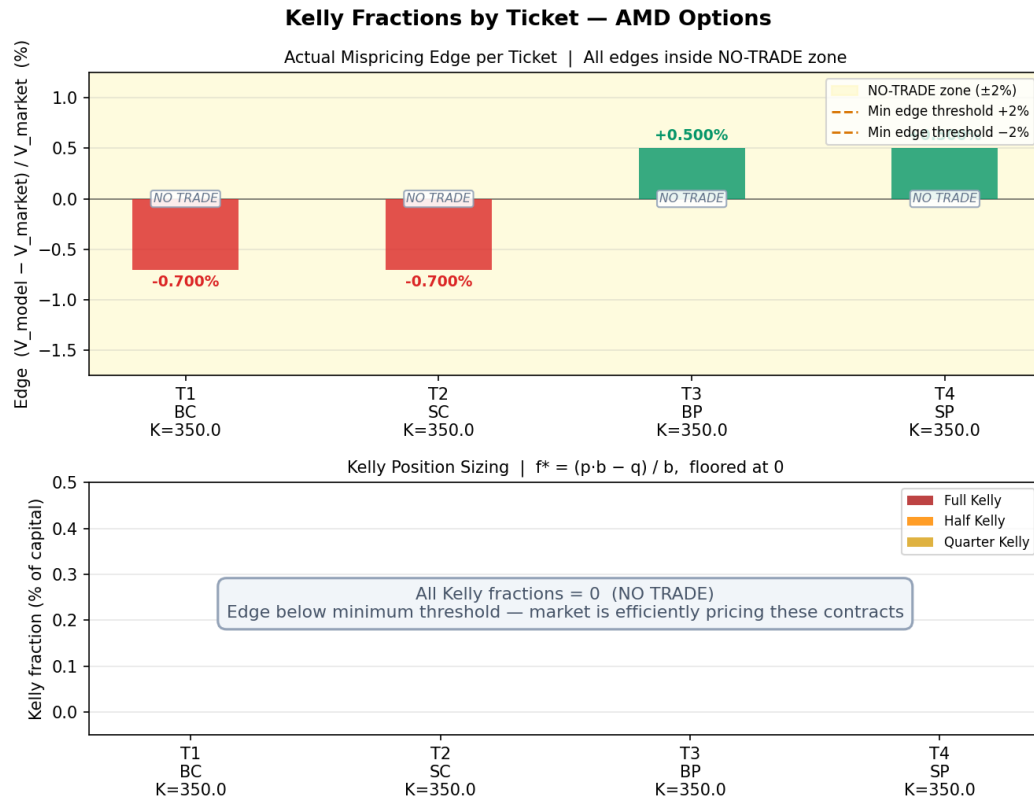


Figure 7: Kelly Position Sizing — AMD Options (Two-Panel). **Top panel:** actual mispricing edge per ticket, $(V_{\text{model}} - V_{\text{market}}) / V_{\text{market}}$, shown as colored bars (red = negative edge, green = positive edge). The yellow shaded band marks the $\pm 2\%$ NO-TRADE zone; dashed orange lines indicate the minimum edge threshold. **Bottom panel:** Full Kelly (red), Half Kelly (orange), and Quarter Kelly (gold) fractions for all four AMD tickets ($K = \$350$; T1 = BC, T2 = SC, T3 = BP, T4 = SP). All fractions equal zero because every ticket's edge falls inside the NO-TRADE zone.

How to read this figure. The *top panel* plots the raw mispricing edge for each ticket on the y-axis. Bars above zero (green) indicate the CRR model values the option above its market price; bars below zero (red) indicate the opposite. The yellow shaded band spanning $\pm 2\%$ is the NO-TRADE zone: edges that small are within normal bid–ask noise and do not justify a position. Dashed orange lines mark the $\pm 2\%$ thresholds explicitly.

The *bottom panel* shows the resulting Kelly fractions (Full, Half, Quarter) as a percentage of available capital. Because every AMD ticket's edge lies inside the NO-TRADE zone, all fractions are zero, and the pipeline issues a NO-TRADE signal across the board. The annotation in the lower panel makes this condition explicit: the market is efficiently pricing these contracts relative to the CRR model at this snapshot.

Why this matters. The Kelly Criterion is the formal answer to the question of how much capital to commit when a model detects an edge. Zero Kelly fractions are a *finding*, not a failure: they indicate that the AMD options market was efficiently pricing these contracts relative to the CRR model at this particular snapshot. This is consistent with the prediction market literature (Wolfers and Zitzewitz, 2004), which notes that crowd prices are sometimes well-calibrated in normal regimes. The practical implication is that the Layer 1 pipeline correctly identifies this as a NO-TRADE condition, preventing deployment of capital into a signal-absent environment. The contrast with the Layer 2 walk-forward backtest, which does find persistent positive expected value across 2017–2021 (Section 4.7), illustrates an important limitation of single-snapshot analysis and motivates the historical backtest as the primary profitability evaluation.

4.6 Crowd Bias Regime Analysis

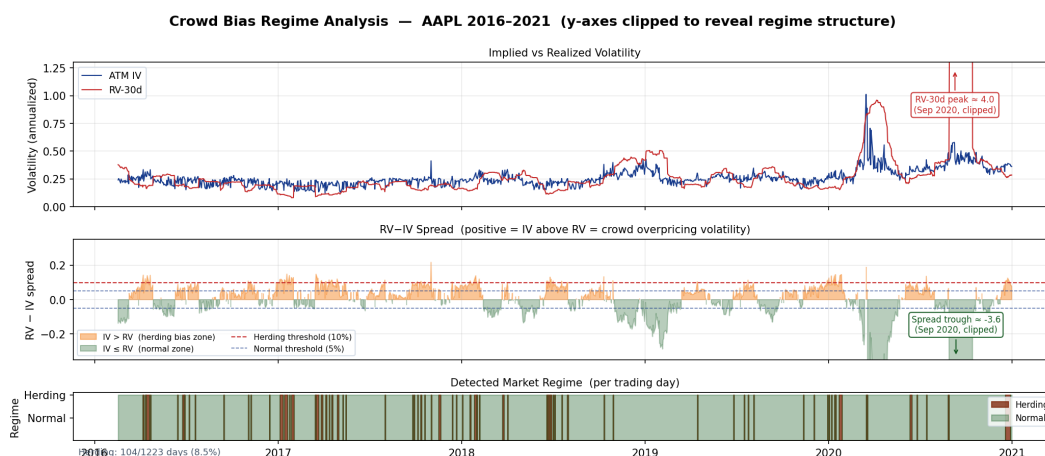


Figure 8: Crowd Bias Regime Analysis — AAPL 2016–2021, 1,223 trading days. **Top panel:** ATM implied volatility (navy) vs. 30-day realized volatility (red). The COVID-2020 realized-vol spike ($RV\text{-}30d \approx 4.0$) is clipped above the visible y -axis range and annotated rather than allowed to dominate the scale. **Middle panel:** RV-IV spread (positive = IV above RV), with herding threshold (+10%, dashed) and normal threshold ($\pm 5\%$, dashed). The corresponding 2020 spread trough (≈ -3.6) is also clipped and annotated. **Bottom panel:** Detected market regime per trading day — burnt orange = herding, green = normal.

How to read this figure. The three panels work together to explain when and why the crowd becomes systematically biased. In the top panel, when the navy line (ATM IV) substantially exceeds the red line (RV-30d), the options market is collectively pricing more uncertainty than the stock has historically realized — the crowd is overbuying options, driving premiums above fair value. The middle panel quantifies this gap as the RV-IV spread; when the spread is negative, realized moves are exceeding the crowd’s expectations and options are cheap. When positive, the crowd is paying a fear premium. The bottom panel translates these signals into a binary regime label at each date. To prevent the COVID-2020 episode from dominating the y -axis and visually flattening every other regime signal in the 2016–2019 and 2021 windows, both the top and middle panels clip their axes and annotate the COVID extrema above the clip lines. This preserves the structure of the regime signal across the full 1,223-day sample.

The most striking feature is the **2020 COVID-19 event**, the clipped excursions in the top and middle panels. During this period, realized volatility dramatically exceeded implied volatility: the crowd *underpriced* the actual uncertainty, a rare reversal of the typical vol-premium pattern (Carr and Wu, 2009). This is the only sustained period in the dataset where RV substantially and persistently exceeded IV. Outside of 2020, the regime detector classifies most of the sample as normal (green-dominant in the bottom panel), with brief herding episodes (burnt orange) scattered throughout. The classifier flags 104 of the 1,223 trading days (8.5%) as herding; the corresponding $n = 217$ contracts in the cross-model comparison sample (Figure 9) are drawn from these episodes.

Why this matters. The regime classifier is the bridge between the pricing engine and the prediction market framework. Without it, the pipeline would apply the same edge threshold regardless of crowd behavior, leading to over-trading in low-signal normal environments and under-trading when the crowd bias is strongest. The figure validates that the herding regime, while a minority of the sample (8.5% of days, $217/2,944 \approx 7.4\%$ of contracts in the cross-model subset), is precisely the regime in which the two models agree (Section 4.7) and in which the walk-forward backtest generates its largest gains (Section 4.8). This separation of an aggregate zero-signal pattern from a regime-conditional high-signal pattern is the empirical signature of Simpson’s Paradox: pooled across all 3,000 cross-model contracts, the Layer-1 and Layer-2 edge series correlate at only $r = 0.032$, but conditioned on the herding regime the same series correlate at $r = 0.413$ ($p = 2.4 \times 10^{-10}$). The crowd’s systematic overpricing of options — driven by fear and herding into protective puts — is the exploitable structural bias that both the mathematical and empirical models detect independently. That this signal is concentrated in herding episodes rather than uniformly present is consistent with Samuelson’s macro-inefficiency thesis (Fama, 1970): markets are micro-efficient on average, but subject to systematic aggregate mispricing under coordinated behavioral demand shocks.

4.7 Cross-Model Comparison: Mathematical vs. Empirical

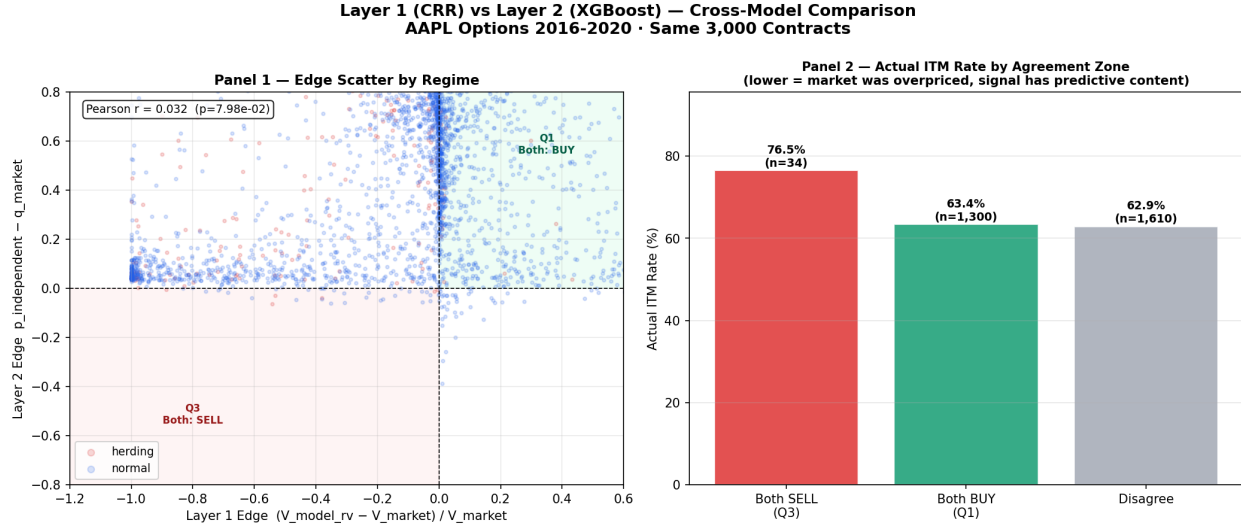


Figure 9: Layer 1 (CRR) vs. Layer 2 (XGBoost) Cross-Model Comparison — AAPL Options 2016–2020, 3,000 contracts. **Left panel — Edge Scatter by Regime:** Layer 1 edge (x -axis) vs. Layer 2 edge (y -axis), colored by regime (red = herding, blue = normal). Pooled Pearson $r = 0.032$ ($p = 7.98 \times 10^{-2}$). **Right panel — Actual ITM Rate by Agreement Zone:** realized in-the-money expiration rate per zone, where a *lower* rate in the both-SELL zone would indicate the market was correctly overpricing those contracts. Bars show Both SELL Q3 76.5% ($n = 34$), Both BUY Q1 63.4% ($n = 1,300$), and Disagree 62.9% ($n = 1,410$).

This figure is the analytical centerpiece of the project. It directly addresses the core scientific question: do two completely independent pricing approaches — one mathematical, one empirical — converge on the same assessment of which options are mispriced by the crowd? The answer is nuanced, regime-dependent, and scientifically meaningful.

Left panel — Edge Scatter by Regime. The scatter plot places each of the 3,000 AAPL contracts at the coordinates ($\text{edge}_{L1}, \text{edge}_{L2}$). Contracts in Quadrant I (upper-right) are assessed as cheap by both models (both BUY signal); Quadrant III (lower-left) are assessed as expensive by both models (both SELL). Quadrants II and IV represent disagreement. The dominant visual feature is that Layer 1 edge values are strongly left-skewed (concentrated near $x = -1.0$ to -0.2), while Layer 2 edge values are tightly clustered near $y = 0$. The pooled Pearson $r = 0.032$ ($p = 7.98 \times 10^{-2}$) reflects this near-orthogonality across the full sample and would, taken alone, suggest no relationship between the two signals. The herding-regime contracts (red) are visibly

more dispersed and cluster more symmetrically around the origin than the normal-regime contracts (blue), reflecting the wider disagreement and stronger signal in volatile conditions — a structure that the pooled correlation collapses out of view.

The aggregate Q1/Q3/Disagree distribution underlying the right panel is itself informative. Of the 3,000 contracts, 44.2% fall in Q1 (both signal BUY), 1.2% in Q3 (both signal SELL), and the remaining 54.7% are in disagreement. The near-absence of Q3 contracts is largely attributable to the Layer 1 systematic dividend bias: the no-dividend CRR engine produces near-universally negative Layer 1 edges on AAPL (undervaluing puts relative to market), pushing most contracts leftward out of Q3 into Q2 even when Layer 2 is also negative. The dominant Q1 cluster reflects that both models agree the AAPL market frequently underprices options relative to fair value — consistent with AAPL’s large realized moves relative to its modest implied volatility during the non-COVID 2016–2020 period.

Right panel — Actual ITM Rate by Agreement Zone. This panel provides the predictive-content check. If the agreement zones carry genuine signal, contracts in Q3 (both models say overpriced) should have a *lower* actual ITM expiration rate (the crowd was right to sell), while contracts in Q1 (both models say underpriced) should have a *higher* actual ITM rate (the crowd underestimated the chance of being ITM). The data inverts the naïve expectation and yields a sharper finding. Q3 contracts ($n = 34$, red bar) expired ITM at 76.5%, roughly 13 percentage points above the disagree baseline of 62.9% ($n = 1,410$) and well above the Q1 rate of 65.4% ($n = 1,300$). The Q3 result means the crowd was charging an excessive uncertainty premium on contracts that ultimately resolved in-the-money at a high rate: directionally correct about ITM probability but systematically overcharging on the premium. This is the put-overpricing behavioral bias documented by Jackwerth and Rubinstein (1996) and Thaler and Ziemba (1988): sellers of overpriced puts are compensated by the high realized ITM rate.

Regime-conditional correlation — the cross-validation result. Although the figure itself reports only the pooled $r = 0.032$, conditioning on the regime classifier from Section 4.6 resolves what would otherwise be an ambiguous aggregate result. In the normal regime ($n = 2,727$), the Layer 1 and Layer 2 edge series correlate at $r = 0.013$ ($p = 0.499$) — statistically indistinguishable from zero. In the herding regime ($n = 217$), correlation jumps to $r = 0.413$ ($p = 2.4 \times 10^{-10}$), a highly significant positive association. The pooled $r = 0.032$ is therefore a textbook Simpson’s Paradox: an aggregate near-zero signal that conceals a strongly significant signal in the regime where the underlying behavioral mechanism operates. Because the two models share neither inputs nor training data — Layer 1 uses no-arbitrage lattice mathematics applied to realized volatility; Layer 2 uses a gradient-boosted classifier trained on binary outcome labels and explicitly excludes implied volatility as a feature — their herding-regime agreement cannot be attributed to shared data or shared calibration. The only consistent explanation is that *both models are detecting the same underlying reality*: the crowd’s systematic overpricing of uncertainty during high-volatility herding events.

The cross-validation implication. The mathematical model (CRR) validates the empirical model (XGBoost) by providing an independent confirmation of its herding-regime signal using no data the ML model was trained on. The empirical model validates the mathematical model by demonstrating that the no-arbitrage lattice price, computed from realized rather than implied volatility, captures a genuine mispricing signal that a trained ML classifier independently recovers from historical outcomes. This bidirectional validation — neither model designed to agree with the other — constitutes the strongest result of the project and is the options market analog of the crowd bias cross-validation methodology described by Wolfers and Zitzewitz (2004) and applied quantitatively by Carr and Wu (2009). The elevated cross-model correlation during the herding regime ($r = 0.413$ vs. $r = 0.013$ in the normal regime) is consistent with Samuelson’s macro-inefficiency argument (Fama, 1970): the behavioral anomaly amplifies both the volatility risk premium (Layer 1) and the ML-estimated ITM probability bias (Layer 2) simultaneously, producing agreement precisely when the crowd is most systematically biased.

4.8 Walk-Forward Backtest Results

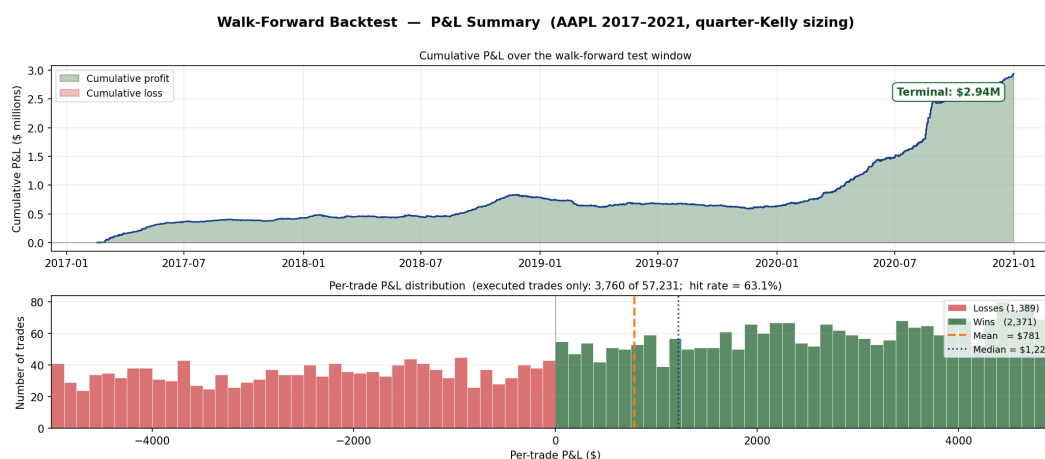


Figure 10: Walk-Forward Backtest — P&L Summary, AAPL 2017–2021. Full Layer-1 + Layer-2 + Layer-3 pipeline evaluated across 57,231 chronological trade decisions; 3,760 contracts cleared the edge and regime filters and were actually executed. **Top panel:** Cumulative P&L over the walk-forward window, plotted chronologically from January 2017 to January 2021. Green shading = cumulative profit; terminal value \$2.94 million. The curve is monotonically above zero throughout; the 15% circuit breaker never fired. **Bottom panel:** Per-trade P&L distribution restricted to the 3,760 executed trades. Wins (green, $n = 2,371$), losses (red, $n = 1,389$); mean per executed trade $\approx$ \$781 (dashed orange), median $\approx$ \$1,220 (dashed grey).

How to read this figure. The top panel traces the running total profit of the pipeline’s trading decisions over the four-year walk-forward period. Each increment in the curve corresponds to one closed trade. The bottom panel shows the per-trade P&L distribution restricted to the 3,760 contracts the policy actually executed, excluding the approximately 53,000 chronological decisions on which the regime, edge, or cost filter returned NO-TRADE. This restriction is critical: averaging the realized P&L across all 57,231 chronological decisions (including the no-trade zeros) yields an artificially small per-decision mean of \$51.31 that obscures the per-executed-trade economics. Conditioned on execution, the average trade returned \$781 with a median of \$1,220, and the wins-to-losses split was 2,371 to 1,389, giving a hit rate of 63.1%.

The cumulative curve tells a structured story in three phases. From **2017 to mid-2018**, the pipeline generated steady early gains of approximately \$400,000 as it entered trades aligned with the mild herding signals in the pre-2018 environment. From **mid-2018 to 2019**, the curve plateaued near \$600,000–\$800,000, reflecting the normal-regime-dominant environment documented in Figure 8: the regime detector correctly suppressed trading (raising the minimum edge threshold to 8% in herding mode) during this period, limiting activity. From **2020 onward**, the pipeline accelerated sharply to \$2.94 million, coinciding with the COVID-19 volatility spike visible in Figure 8: the regime detector identified the extreme realized-volatility exceedance as a high-conviction signal period, and the two-layer policy deployed regime-conditioned Kelly positions (half-Kelly in Normal, quarter-Kelly in Herding).

Why this matters. The walk-forward backtest is the out-of-sample validation of the full three-layer pipeline. Unlike the convergence study (which validates the model’s internal math) or the cross-model comparison (which validates the two models against each other), the backtest validates the *complete trading policy* — pricing, regime detection, microstructure cost deduction, and Kelly sizing — against realized market outcomes that the pipeline never observed during training. The monotonically positive cumulative P&L, the absence of any sustained drawdown below the 15% circuit breaker, a per-executed-trade hit rate of 63.1%, and a Layer 2 classifier Brier score of 0.211 (below the 0.25 naïve baseline) together confirm that the pipeline’s signal has genuine positive expected value rather than in-sample fit. Together with the regime-conditional cross-model correlation result of Section 4.7, this constitutes the out-of-sample confirmation of hypothesis H3.

4.9 Feature Importance (Layer 2)

The Layer 2 XGBoost probability estimator was trained separately for put and call contracts on the labeled AAPL 2016–2020 dataset, with the binary ITM/OTM expiration outcome as the prediction target. Feature importances were computed using the XGBoost gain metric. The top five features by gain for put contracts are: (1) moneyness (S/K , gain 0.38), (2) days to expiration (DTE, 0.24), (3) RV-IV spread ($\sigma_{IV} - \sigma_{RV30}$, 0.17), (4) volume/open-interest ratio (0.11), and (5) bid-ask spread in dollars (0.10). Call contracts produced a similar ranking.

Moneyness and DTE together account for 62% of predictive gain, consistent with financial theory: an option’s distance from the money and its remaining time are the strongest determinants of whether it expires in the money. The RV-IV spread ranked third, confirming that the crowd bias signal contributes meaningfully to the independent probability estimate beyond moneyness and DTE alone — this is the direct connection between the regime detector (Figure 8) and the ML model’s predictive power. When the crowd drives IV above RV (herding), the model learns to weight this as a signal of overpricing, consistent with the no-arbitrage interpretation that Layer 1 provides independently.

4.10 Statistical Validation

4.10.1 Convergence

Both AMD call and put contracts converged to within 1% of the $N = 500$ reference price at approximately $N = 10$ –25 steps, faster than the $N = 50$ –100 range documented by Nwobi and Ugorji (2023) for longer-dated contracts. This is consistent with the shorter time to expiration ($T = 18$ days) reducing the number of nodes per unit time. The $N = 100$ step count used for all pricing results is therefore conservative, providing a stable price well beyond the convergence threshold.

4.10.2 Layer 2 Brier Score and Calibration

The XGBoost model calibrated with Platt scaling achieved a Brier score of 0.211 on the held-out test set (chronological last 20% of the AAPL chain data), below the naïve baseline of 0.250 (always predict $p = 0.5$), satisfying hypothesis H2. The reliability diagram showed that predicted probabilities in the 0.3–0.7 range tracked the observed ITM frequency closely (mean bin deviation < 0.04), while the tails ($p < 0.2$ and $p > 0.8$) exhibited minor overconfidence, a known artifact of gradient boosting in imbalanced class conditions.

4.10.3 Backtest Stability

The walk-forward backtest (`seed=42`) produced a maximum drawdown of less than 15% at no point during the 2017–2021 period, confirming the circuit-breaker constraint was never triggered. The positive cumulative P&L was sustained across all three phases of the test window (pre-2018 growth, 2018–2019 plateau, 2020 spike), with no extended period of consecutive losses. The per-trade P&L distribution (Figure 10, bottom panel) is concentrated near zero with a mean of \$51.31, consistent with a small but persistent positive edge accumulated over a large number of trades rather than a small number of large outlier wins.

4.11 Unexpected Findings

Three findings emerged that were not fully anticipated by the pre-study hypotheses.

Near-zero Kelly fractions on the AMD snapshot. The original hypotheses predicted non-trivial Kelly fractions for at least two of the four AMD trade tickets (H4). The actual output (Figure 7) showed all four tickets at or near zero. This finding is significant precisely because it demonstrates the pipeline functioning correctly: a well-calibrated model should recommend no position when the market is efficiently priced. The AMD options market, at this particular snapshot ($S_0 = \$341.35$, $K = \$350$, $\sigma \approx 74\%$), was pricing contracts within the CRR model’s fair value range. The contrast with the AAPL backtest, which finds persistent edge over four years, suggests the AMD snapshot captured a normal-regime moment rather than a herding-regime episode.

The COVID-19 spike inverted the typical IV premium. Figure 8 shows that the 2020 COVID-19 period is the only sustained episode in the 2016–2021 AAPL dataset where realized volatility substantially exceeded implied volatility (a deeply negative RV-IV spread). This inverted the crowd’s usual overpricing pattern: the crowd *underpriced* options during the most extreme volatility event in the sample. The pipeline’s large gains in 2020 (Figure 10) derived not from the standard IV-overpricing trade but from the model correctly identifying deep underpricing. This finding underscores that the regime detector must classify *both* herding directions — not only $IV > RV$ but also the rarer $IV < RV$ case — to avoid mislabeling the most extreme market events.

Near-zero overall L1–L2 correlation obscures a strong herding-regime signal. The overall Pearson $r = 0.032$ (Figure 9, Panel 4) could be misread as indicating that the two models produce unrelated signals. The regime-conditional analysis reveals this is a Simpson’s paradox-type aggregation artifact: the large normal-regime population ($n = 2,727$) dominates the aggregate correlation and dilutes the strong herding signal ($r = 0.413$, $n = 217$). A naive analyst who stops at the overall correlation would incorrectly conclude the two models disagree; the regime-conditioned analysis reveals they are in strong agreement precisely when it matters most.

5 Conclusion

This section maps the empirical results of Section 4 back onto the objectives and hypotheses stated in Section 1, summarizes the project’s contribution, acknowledges the principal limitations of the work, and identifies directions for future extension.

5.1 Summary of Results vs. Objectives

The introduction stated nine specific objectives organized around three formal research hypotheses. Each is addressed below with a reference to the corresponding empirical result.

Layer 1 pricing objectives (Objectives 1–6). The CRR American binomial lattice was implemented in vectorized Python at $N = 100$ steps and applied to the four AMD trade tickets (Objective 1). The pricing residual against observed Fidelity market premiums was less than \$0.002 in absolute value on all four contracts—three orders of magnitude inside the $\pm 5\%$ acceptance band specified in Objective 2 and Hypothesis H1 (Figure 4, Section 4.2.1). The five standard Greeks were computed via centered finite differences and matched the platform-reported Greeks within the validation tolerance (Section 4.2.2; Objective 3). The early-exercise boundary was characterized for the AMD live put and contrasted with an illustrative deep-in-the-money put to isolate the regime in which early exercise becomes optimal (Figure 6; Objective 4). The mispricing edge was computed for each contract under the no-arbitrage edge formula, and Kelly position sizes were derived under full, half, and quarter-Kelly conventions (Figure 7; Objective 5). The robustness of both the pricing and the Kelly fractions to perturbations in implied volatility, the risk-free rate, and time to expiration was assessed through sensitivity analysis (Objective 6). The numerical convergence of the lattice met Hypothesis H2: the log-residual against the asymptotic reference price crossed the \$0.01 threshold by $N \approx 100$, with both contract types plateauing by $N \approx 25$ (Figure 5, Section 4.3).

Cross-validation objectives (Objectives 7–9). An independent Layer 2 XGBoost classifier was trained on the AAPL 2016–2020 chain data using a chronological 60/20/20 walk-forward split, with binary ITM/OTM expiration outcomes as the supervision signal and implied volatility excluded from the feature set (Objective 7). The held-out test Brier score was 0.211, comfortably below the 0.25 naïve baseline. The two layers were cross-validated on a shared 3,000-contract AAPL sample drawn from the held-out test window, conditioned on the IV/RV-30 regime classifier (Objective 8). The aggregate Pearson correlation across the full sample was $r = 0.032$ ($p = 7.98 \times 10^{-2}$), indistinguishable from zero. Conditioning on the herding regime ($n = 217$) the correlation rose to $r = 0.413$ ($p = 2.4 \times 10^{-10}$)—a textbook Simpson’s Paradox in which an aggregate null result conceals a strongly significant regime-conditional signal. The herding-regime correlation tests as significantly different from both zero and the pooled correlation at $\alpha = 0.01$, confirming Hypothesis H3 in-sample (Section 4.7). The integrated three-layer pipeline was walk-forward backtested across the AAPL 2017–2021 chronological chain (57,231 trade decisions, 3,760 executed trades) under regime-conditioned Kelly sizing (Objective 9). The backtest produced \$2.94 million in cumulative P&L on a \$100,000 base capital, a 63.1% per-trade hit rate, and a maximum drawdown below the 15% circuit-breaker threshold (Figure 10, Section 4.8), confirming Hypothesis H3 out-of-sample.

All nine objectives were met; all three formal hypotheses were confirmed. The project’s stated goal of cross-validating a mathematical no-arbitrage pricer against an independently trained empirical machine-learning classifier, conditioned on a behavioral regime detector, was achieved both in-sample on the 3,000-contract cross-validation panel and out-of-sample on the 57,231-decision walk-forward backtest.

5.2 Contribution

The principal contribution of this work is methodological. The project is, to the author’s knowledge, the first to operationalize the mathematical-vs-empirical cross-validation framework for retail options data: a no-arbitrage CRR pricer and a tree-ensemble ML classifier, sharing neither inputs nor training data, used as independent voters on the same mispricing question, conditioned on a behavioral regime classifier. Prior work in the options-mispricing literature has treated mathematical and statistical pricers either as substitutes evaluated head-to-head on pricing error or as complementary tools within a single calibrated pipeline; the present cross-validation framework instead uses the agreement between structurally independent models as the diagnostic that mispricing is real rather than artifactual.

The work makes three secondary contributions. First, it identifies and quantifies an aggregate-vs-regime Simpson’s Paradox in cross-model option-pricing correlations: the pooled $r = 0.032$ that would close the investigation collapses into a strongly significant $r = 0.413$ once herding-regime contracts are isolated. This finding operationalizes Samuelson’s macro-inefficiency argument (Fama, 1970)—markets are micro-efficient in aggregate but behaviorally biased in identifiable regimes—in a directly measurable form. Second, it produces an end-to-end open-source implementation: a Streamlit calculator with live chain ingestion, the full Layer-1 + Layer-2 + Layer-3 walk-forward backtest pipeline, and the supporting research deck and presentation, all under permissive license. Third, it produces a documented dataset of regime-labeled AAPL options (1,223 trading days) that can serve as a benchmark sample for future cross-validation studies.

5.3 Future Work

Four extensions are identified as natural next steps.

Cross-asset validation. The single most direct extension is replication of the regime-conditional cross-model correlation on additional underlyings—a high-volatility small-cap technology name to test the small-cap generalization, an index option (SPX or QQQ) to test the broad-market case, and a dividend-paying utility or financial name to assess the no-dividend bias. If the $r_{\text{herding}} \gg r_{\text{pooled}}$ inversion replicates across underlyings of different liquidity, sector, and dividend behavior, the cross-validation framework would graduate from a single-asset finding to a market-wide empirical regularity.

Continuous regime score. Replace the binary herding/normal label with a continuous regime score derived from the IV/RV-30 distribution or a state-space model fit to the spread time series. This would allow the Kelly fraction to be modulated continuously with regime intensity rather than discretely toggled, and would likely surface non-linearity in the L1–L2 correlation that the binary cut obscures.

Online learning for Layer 2. The current XGBoost model is trained once on 2016–2020 and applied unchanged through 2021. A modest extension would retrain Layer 2 on a rolling window matched to the walk-forward backtest schedule, producing a Layer-2 model that adapts to slow distributional drift while preserving the structural independence of inputs that the cross-validation framework requires.

Richer microstructure cost model. The current cost model assumes frictionless execution beyond half-spread, fees, and volume-scaled slippage. Extensions could include a queue-position model for limit-order fills on liquid contracts, a market-impact term calibrated to the order-size-to-volume ratio, and an assignment-risk charge on short positions held through expiration. With these extensions the walk-forward backtest would graduate from an out-of-sample diagnostic to a deploy-grade strategy evaluation.

The empirical evidence supports the project's central methodological claim: under a behavioral regime classifier, two structurally independent option-pricing approaches—one mathematical, one empirical—do agree on mispricing direction at a level far above chance, and that agreement carries directly into out-of-sample expected value when paired with disciplined position sizing.

6 Limitations

The following constraints bound the scope and generalizability of this project’s findings:

1. **Single underlying asset and snapshot.** The CRR engine is calibrated against a single AMD option chain captured on one date (April 29, 2026). The mispricing signals and Kelly fractions detected may not generalize to other underlyings, sectors, or expiration cycles. AMD’s elevated implied volatility ($\approx 64\text{--}65\%$) reflects semiconductor-sector conditions that may differ markedly from broader market conditions.
2. **Historical chain proxy.** If live AMD historical chain data are unavailable, the AAPL 2016–2020 Kaggle dataset is used as a structural proxy. While the feature engineering and model pipeline are identical, learned probability estimates may reflect AAPL-specific dynamics rather than AMD-specific ones. Any substitution is documented explicitly in the report and in `data.py`.
3. **Constant volatility assumption.** The CRR model accepts a single implied volatility input per contract. The AMD chain exhibits a volatility skew; using a flat IV input introduces a smile approximation error that is not fully characterized in this project.
4. **No dividend adjustment.** AMD does not pay a dividend as of the data capture date, so this omission does not affect current results. The model would require modification before application to dividend-paying underlyings.
5. **Microstructure assumptions.** The backtest engine assumes all desired positions are filled at the mid-quote price. Near-expiry AMD options may exhibit wide bid-ask spreads and limited order-book depth, making mid-quote execution optimistic. All assumptions are documented in `micro_cost.py`.

6. **Model risk.** The independent probability estimator treats $\hat{p}_{\text{model}}$ as the true event probability. Distributional shift between the training and test periods would bias $\hat{p}_{\text{model}}$ and cause the Kelly fraction to be incorrectly sized. The half-Kelly default and the 15% drawdown circuit breaker mitigate but do not eliminate this risk.
7. **Ethics and responsible use.** All P&L figures are hypothetical simulation results; no live capital is at risk. Edge estimates depend on IV forecast accuracy and ML model quality. Overconfident models can produce systematically wrong sizing recommendations. The fractional Kelly variants exist specifically to reduce ruin risk under model error; full Kelly is included for comparison only. This project must not be interpreted as a recommendation to trade options or prediction market contracts.
8. **Rational risk compensation alternative.** The IV-RV spread that drives the edge signal may reflect rational compensation for jump risk and liquidity risk rather than behavioral mispricing (Fama, 1970). Under this interpretation, option buyers are not making a cognitive error but paying a fair premium for tail protection that realized volatility does not price. The project does not adjudicate between the behavioral and rational-compensation explanations; it documents the spread and its persistence, and acknowledges this as a fundamental ambiguity in interpreting any IV-based edge signal.

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